DISCO-TAB — Comprehensive Benchmark

all generators × 24 datasets × all metrics; cells are **mean ± 95% confidence interval** over random seeds (Student-*t*, up to 3 seeds); **bold** = best per row (by mean, where a direction is defined); – = missing run/metric; ± absent where only one seed is available.

Utility (downstream TSTR)

Table 1: TSTR F1-macro (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.783 ± 0.184	0.483 ± 0.237	0.135 ± 0.582	0.591 ± 0.252	0.406	0.635 ± 0.151	0.396 ± 0.174	0.600 ± 0.270	0.559 ± 0.371	0.394 ± 0.217	0.388	**0.795**	0.787
Breast Cancer	0.930 ± 0.035	0.295 ± 0.304	0.934 ± 0.084	0.912 ± 0.041	0.924	0.934 ± 0.044	0.497 ± 0.461	**0.937 ± 0.047**	0.855 ± 0.082	0.513 ± 0.508	–	–	–
Liver Disorders	0.509 ± 0.208	0.473 ± 0.224	0.527 ± 0.171	0.553 ± 0.098	**0.592**	0.542 ± 0.174	0.488 ± 0.263	0.490 ± 0.333	0.500 ± 0.106	0.380 ± 0.208	–	–	–
Parkinsons	0.646 ± 0.517	0.480 ± 0.344	0.000 ± 0.000	0.630 ± 0.104	**0.928**	0.737 ± 0.215	0.424 ± 0.012	0.704 ± 0.131	0.695 ± 0.096	0.446 ± 0.139	–	–	–
Obesity	**0.886 ± 0.011**	0.138 ± 0.161	0.490 ± 0.086	0.243 ± 0.045	0.795	0.655 ± 0.097	0.115 ± 0.033	0.607 ± 0.092	0.171 ± 0.091	0.102 ± 0.026	–	–	0.637
Diabetes	0.843 ± 0.186	0.836 ± 0.042	0.875 ± 0.003	0.485 ± 0.014	**0.892 ± 0.006**	0.797 ± 0.086	0.478 ± 0.000	0.727 ± 0.121	0.658 ± 0.070	0.377 ± 0.082	0.517	–	0.862
German Credit	0.516 ± 0.572	0.448 ± 0.046	0.137 ± 0.591	0.496 ± 0.118	0.454 ± 0.107	**0.594 ± 0.194**	0.468 ± 0.203	0.501 ± 0.164	0.449 ± 0.145	0.459 ± 0.077	–	–	–
Bank Marketing	**0.673 ± 0.205**	0.604 ± 0.039	0.673 ± 0.043	0.469 ± 0.001	0.637 ± 0.018	0.608 ± 0.082	0.469 ± 0.000	0.498 ± 0.058	0.471 ± 0.005	0.236 ± 0.317	–	–	–
Census Income	**0.765 ± 0.039**	0.727 ± 0.031	0.746 ± 0.014	0.459 ± 0.038	0.750 ± 0.020	0.669 ± 0.126	0.433 ± 0.002	0.549 ± 0.126	0.615 ± 0.145	0.524 ± 0.264	–	–	–
Melanoma	0.536 ± 0.169	0.406 ± 0.142	0.000 ± 0.000	**0.606 ± 0.185**	0.510 ± 0.063	0.557 ± 0.099	0.442 ± 0.070	0.529 ± 0.128	0.508 ± 0.201	0.445 ± 0.588	–	–	–
PBC	0.522 ± 0.515	0.490 ± 0.120	0.376 ± 0.000	0.534 ± 0.108	0.564 ± 0.019	0.590 ± 0.152	0.492 ± 0.182	**0.646 ± 0.177**	0.566 ± 0.220	0.428 ± 0.123	–	–	–
PBC3	0.468 ± 0.398	0.475 ± 0.105	0.000 ± 0.000	0.530 ± 0.014	0.453 ± 0.000	**0.632 ± 0.095**	0.453 ± 0.000	0.469 ± 0.098	0.477 ± 0.101	0.497 ± 0.131	0.453	–	0.617
Retinopathy	0.518 ± 0.146	0.509 ± 0.257	0.124 ± 0.535	0.484 ± 0.040	0.452 ± 0.101	**0.539 ± 0.211**	0.480 ± 0.201	0.478 ± 0.147	0.513 ± 0.146	0.511 ± 0.042	–	–	–
WHAS500	0.739 ± 0.032	0.428 ± 0.040	0.476 ± 0.227	0.670 ± 0.213	**0.763 ± 0.059**	0.699 ± 0.159	0.395 ± 0.161	0.619 ± 0.114	0.709 ± 0.046	0.324 ± 0.096	–	–	0.643
GBSG2	0.580 ± 0.067	0.438 ± 0.073	0.505 ± 0.182	0.526 ± 0.181	0.547 ± 0.135	0.538 ± 0.086	0.474 ± 0.204	0.556 ± 0.082	**0.590 ± 0.196**	0.475 ± 0.310	0.460	–	–
Colon	**0.587 ± 0.120**	0.506 ± 0.146	0.534 ± 0.031	0.541 ± 0.033	0.577 ± 0.063	0.497 ± 0.250	0.426 ± 0.203	0.564 ± 0.075	0.526 ± 0.083	0.450 ± 0.097	–	–	–
GRACE	0.609 ± 0.052	0.456 ± 0.103	0.437 ± 0.083	**0.610 ± 0.067**	0.576 ± 0.068	0.591 ± 0.100	0.456 ± 0.074	0.508 ± 0.058	0.543 ± 0.224	0.463 ± 0.033	–	–	–
TRACE	**0.676 ± 0.019**	0.513 ± 0.077	0.666 ± 0.081	0.624 ± 0.028	0.660 ± 0.121	0.646 ± 0.163	0.445 ± 0.065	0.516 ± 0.148	0.609 ± 0.192	0.516 ± 0.045	–	–	–
Framingham	**0.581 ± 0.088**	0.577 ± 0.090	0.455 ± 0.046	0.488 ± 0.036	0.556 ± 0.031	0.546 ± 0.088	0.474 ± 0.112	0.486 ± 0.104	0.453 ± 0.122	0.448 ± 0.110	–	–	–
Blood Transfusion	0.351 ± 0.755	0.481 ± 0.151	0.000 ± 0.000	**0.579 ± 0.146**	0.541 ± 0.065	0.498 ± 0.141	0.510 ± 0.243	0.517 ± 0.078	0.456 ± 0.215	0.502 ± 0.198	–	–	–
Credit-G	**0.643 ± 0.103**	0.421 ± 0.034	0.412 ± 0.000	0.476 ± 0.143	0.491 ± 0.014	0.577 ± 0.077	0.435 ± 0.099	0.465 ± 0.128	0.448 ± 0.085	0.434 ± 0.200	–	–	–
QSAR Biodeg	**0.796 ± 0.129**	0.419 ± 0.028	0.662 ± 0.126	0.712 ± 0.106	0.391 ± 0.025	0.729 ± 0.081	0.428 ± 0.232	0.770 ± 0.071	0.612 ± 0.298	0.393 ± 0.338	–	–	–
Maternal Risk	**0.624 ± 0.257**	0.305 ± 0.042	0.000 ± 0.000	0.375 ± 0.160	0.296 ± 0.087	0.463 ± 0.240	0.235 ± 0.187	0.527 ± 0.099	0.395 ± 0.082	0.234 ± 0.260	–	–	0.549
Website Phishing	**0.614 ± 0.075**	0.329 ± 0.130	0.521 ± 0.049	0.359 ± 0.081	0.604 ± 0.064	0.555 ± 0.090	0.263 ± 0.217	0.492 ± 0.047	0.454 ± 0.013	0.286 ± 0.087	–	–	–

Table 2: TSTR F1-weighted (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.799 ± 0.195	0.559 ± 0.243	0.185 ± 0.796	0.679 ± 0.178	0.555	0.710 ± 0.122	0.500 ± 0.317	0.676 ± 0.173	0.655 ± 0.256	0.416 ± 0.438	0.530	**0.827**	0.823
Breast Cancer	0.935 ± 0.033	0.284 ± 0.354	0.939 ± 0.079	0.920 ± 0.038	0.929	0.939 ± 0.042	0.489 ± 0.601	**0.942 ± 0.045**	0.869 ± 0.068	0.558 ± 0.428	–	–	–
Liver Disorders	0.511 ± 0.205	0.472 ± 0.228	0.529 ± 0.169	0.553 ± 0.099	**0.592**	0.543 ± 0.175	0.488 ± 0.265	0.490 ± 0.333	0.501 ± 0.105	0.384 ± 0.206	–	–	–
Parkinsons	0.732 ± 0.378	0.632 ± 0.164	0.000 ± 0.000	0.729 ± 0.116	**0.947**	0.801 ± 0.156	0.630 ± 0.018	0.784 ± 0.111	0.768 ± 0.100	0.584 ± 0.126	–	–	–
Obesity	**0.888 ± 0.010**	0.138 ± 0.168	0.493 ± 0.088	0.250 ± 0.046	0.796	0.655 ± 0.108	0.120 ± 0.036	0.610 ± 0.098	0.176 ± 0.096	0.101 ± 0.023	–	–	0.636
Diabetes	0.956 ± 0.045	0.952 ± 0.017	0.962 ± 0.001	0.876 ± 0.003	**0.968 ± 0.002**	0.945 ± 0.021	0.874 ± 0.001	0.927 ± 0.029	0.901 ± 0.039	0.584 ± 0.144	0.881	–	0.961
German Credit	0.502 ± 0.722	0.585 ± 0.036	0.192 ± 0.827	0.624 ± 0.074	0.604 ± 0.067	**0.680 ± 0.112**	0.601 ± 0.099	0.631 ± 0.105	0.587 ± 0.049	0.516 ± 0.219	–	–	–
Bank Marketing	0.865 ± 0.036	0.846 ± 0.049	0.854 ± 0.030	0.828 ± 0.000	**0.872 ± 0.004**	0.862 ± 0.019	0.828 ± 0.000	0.835 ± 0.015	0.829 ± 0.001	0.273 ± 0.618	–	–	–
Census Income	**0.830 ± 0.014**	0.812 ± 0.018	0.818 ± 0.016	0.671 ± 0.019	0.826 ± 0.013	0.782 ± 0.067	0.656 ± 0.002	0.718 ± 0.069	0.748 ± 0.076	0.672 ± 0.164	–	–	–
Melanoma	0.561 ± 0.205	0.573 ± 0.124	0.000 ± 0.000	**0.706 ± 0.136**	0.660 ± 0.041	0.676 ± 0.045	0.588 ± 0.059	0.645 ± 0.130	0.614 ± 0.026	0.454 ± 0.688	–	–	–
PBC	0.507 ± 0.610	0.502 ± 0.119	0.454 ± 0.000	0.544 ± 0.102	0.584 ± 0.026	0.592 ± 0.167	0.528 ± 0.205	**0.662 ± 0.191**	0.586 ± 0.156	0.444 ± 0.124	–	–	–
PBC3	0.541 ± 0.652	0.756 ± 0.039	0.000 ± 0.000	0.780 ± 0.013	0.751 ± 0.000	0.786 ± 0.079	0.751 ± 0.000	0.749 ± 0.037	0.748 ± 0.012	0.675 ± 0.082	0.751	–	**0.804**
Retinopathy	0.542 ± 0.133	0.526 ± 0.279	0.148 ± 0.636	0.515 ± 0.040	0.480 ± 0.099	**0.565 ± 0.183**	0.481 ± 0.265	0.507 ± 0.136	0.531 ± 0.148	0.517 ± 0.032	–	–	–
WHAS500	0.752 ± 0.024	0.493 ± 0.047	0.538 ± 0.159	0.694 ± 0.198	**0.779 ± 0.055**	0.719 ± 0.127	0.428 ± 0.101	0.642 ± 0.126	0.721 ± 0.042	0.339 ± 0.098	–	–	0.668
GBSG2	0.587 ± 0.063	0.461 ± 0.045	0.530 ± 0.145	0.539 ± 0.168	0.558 ± 0.123	0.542 ± 0.110	0.486 ± 0.213	0.564 ± 0.073	**0.596 ± 0.186**	0.474 ± 0.344	0.477	–	–
Colon	**0.586 ± 0.122**	0.506 ± 0.148	0.534 ± 0.033	0.541 ± 0.032	0.577 ± 0.062	0.495 ± 0.255	0.429 ± 0.200	0.565 ± 0.074	0.526 ± 0.084	0.451 ± 0.092	–	–	–
GRACE	0.639 ± 0.066	0.551 ± 0.075	0.566 ± 0.057	**0.668 ± 0.066**	0.651 ± 0.047	0.656 ± 0.050	0.526 ± 0.150	0.591 ± 0.033	0.634 ± 0.153	0.503 ± 0.096	–	–	–
TRACE	**0.676 ± 0.018**	0.513 ± 0.075	0.665 ± 0.082	0.624 ± 0.029	0.660 ± 0.120	0.645 ± 0.166	0.444 ± 0.069	0.516 ± 0.147	0.609 ± 0.191	0.516 ± 0.047	–	–	–
Framingham	0.622 ± 0.118	**0.641 ± 0.084**	0.586 ± 0.025	0.606 ± 0.025	0.639 ± 0.022	0.635 ± 0.054	0.590 ± 0.057	0.580 ± 0.101	0.580 ± 0.057	0.490 ± 0.119	–	–	–
Blood Transfusion	0.417 ± 0.898	0.520 ± 0.232	0.000 ± 0.000	**0.676 ± 0.122**	0.647 ± 0.058	0.607 ± 0.048	0.636 ± 0.128	0.615 ± 0.105	0.590 ± 0.192	0.551 ± 0.266	–	–	–
Credit-G	**0.675 ± 0.121**	0.577 ± 0.018	0.576 ± 0.000	0.618 ± 0.097	0.617 ± 0.004	0.657 ± 0.038	0.582 ± 0.025	0.606 ± 0.075	0.588 ± 0.063	0.501 ± 0.385	–	–	–
QSAR Biodeg	**0.819 ± 0.106**	0.541 ± 0.016	0.709 ± 0.105	0.755 ± 0.090	0.519 ± 0.034	0.767 ± 0.066	0.465 ± 0.314	0.799 ± 0.059	0.674 ± 0.210	0.361 ± 0.442	–	–	–
Maternal Risk	**0.660 ± 0.234**	0.340 ± 0.080	0.000 ± 0.000	0.447 ± 0.165	0.386 ± 0.041	0.510 ± 0.333	0.306 ± 0.153	0.592 ± 0.068	0.485 ± 0.087	0.251 ± 0.394	–	–	0.622
Website Phishing	**0.746 ± 0.068**	0.419 ± 0.082	0.717 ± 0.068	0.497 ± 0.115	0.737 ± 0.041	0.694 ± 0.052	0.346 ± 0.299	0.662 ± 0.035	0.614 ± 0.058	0.368 ± 0.100	–	–	–

Table 3: TSTR accuracy (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.794 ± 0.204	0.572 ± 0.305	0.228 ± 0.980	0.733 ± 0.110	0.683	0.750 ± 0.110	0.572 ± 0.514	0.711 ± 0.086	0.717 ± 0.143	0.472 ± 0.335	0.633	**0.833**	**0.833**
Breast Cancer	0.936 ± 0.033	0.316 ± 0.278	0.939 ± 0.079	0.921 ± 0.038	0.930	0.939 ± 0.044	0.561 ± 0.437	**0.942 ± 0.045**	0.874 ± 0.055	0.579 ± 0.363	–	–	–
Liver Disorders	0.546 ± 0.150	0.502 ± 0.181	0.551 ± 0.144	0.556 ± 0.091	**0.594**	0.551 ± 0.191	0.493 ± 0.252	0.493 ± 0.330	0.507 ± 0.095	0.473 ± 0.162	–	–	–
Parkinsons	0.735 ± 0.368	0.675 ± 0.074	0.000 ± 0.000	0.744 ± 0.169	**0.949**	0.803 ± 0.147	0.735 ± 0.037	0.795 ± 0.127	0.769 ± 0.127	0.607 ± 0.224	–	–	–
Obesity	**0.887 ± 0.012**	0.154 ± 0.127	0.499 ± 0.077	0.297 ± 0.056	0.801	0.655 ± 0.120	0.149 ± 0.042	0.627 ± 0.083	0.270 ± 0.101	0.115 ± 0.024	–	–	0.643
Diabetes	0.961 ± 0.031	0.954 ± 0.022	0.963 ± 0.000	0.915 ± 0.001	**0.970 ± 0.002**	0.953 ± 0.015	0.915 ± 0.001	0.941 ± 0.020	0.909 ± 0.095	0.479 ± 0.151	0.916	–	0.965
German Credit	0.542 ± 0.498	0.663 ± 0.147	0.233 ± 1.004	0.700 ± 0.033	0.710 ± 0.022	0.712 ± 0.040	0.680 ± 0.045	**0.715 ± 0.054**	0.668 ± 0.094	0.545 ± 0.255	–	–	–
Bank Marketing	0.867 ± 0.064	0.858 ± 0.094	0.845 ± 0.072	0.883 ± 0.000	**0.896 ± 0.002**	0.889 ± 0.003	0.883 ± 0.000	0.884 ± 0.004	0.883 ± 0.000	0.266 ± 0.436	–	–	–
Census Income	0.832 ± 0.004	0.823 ± 0.016	0.821 ± 0.022	0.766 ± 0.005	**0.835 ± 0.012**	0.812 ± 0.032	0.758 ± 0.007	0.782 ± 0.034	0.784 ± 0.036	0.698 ± 0.183	–	–	–
Melanoma	0.545 ± 0.185	0.626 ± 0.152	0.000 ± 0.000	**0.724 ± 0.126**	**0.724 ± 0.070**	0.707 ± 0.000	0.626 ± 0.185	0.667 ± 0.195	0.634 ± 0.160	0.463 ± 0.655	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
PBC	0.566 ± 0.373	0.497 ± 0.120	0.603 ± 0.000	0.540 ± 0.104	0.587 ± 0.039	0.593 ± 0.159	0.561 ± 0.277	**0.667 ± 0.209**	0.603 ± 0.079	0.439 ± 0.127	–	–	–
PBC3	0.543 ± 0.663	0.824 ± 0.020	0.000 ± 0.000	**0.838 ± 0.020**	0.829 ± 0.000	0.781 ± 0.102	0.829 ± 0.000	0.810 ± 0.020	0.805 ± 0.102	0.652 ± 0.164	0.829	–	0.829
Retinopathy	0.554 ± 0.121	0.532 ± 0.301	0.198 ± 0.853	0.541 ± 0.089	0.491 ± 0.103	**0.581 ± 0.146**	0.514 ± 0.220	0.527 ± 0.146	0.532 ± 0.155	0.514 ± 0.034	–	–	–
WHAS500	0.753 ± 0.027	0.573 ± 0.108	0.627 ± 0.015	0.706 ± 0.193	**0.785 ± 0.053**	0.728 ± 0.086	0.452 ± 0.080	0.649 ± 0.147	0.720 ± 0.046	0.333 ± 0.093	–	–	0.677
GBSG2	0.589 ± 0.068	0.505 ± 0.068	0.589 ± 0.063	0.548 ± 0.146	0.563 ± 0.105	0.560 ± 0.153	0.493 ± 0.230	0.565 ± 0.065	**0.597 ± 0.180**	0.488 ± 0.280	0.493	–	–
Colon	**0.593 ± 0.098**	0.508 ± 0.149	0.542 ± 0.014	0.542 ± 0.028	0.578 ± 0.065	0.550 ± 0.089	0.492 ± 0.193	0.569 ± 0.071	0.535 ± 0.066	0.475 ± 0.056	–	–	–
GRACE	0.630 ± 0.069	0.592 ± 0.050	0.680 ± 0.022	0.680 ± 0.078	0.685 ± 0.033	0.678 ± 0.029	0.545 ± 0.228	0.625 ± 0.062	**0.692 ± 0.072**	0.495 ± 0.097	–	–	–
TRACE	**0.679 ± 0.017**	0.516 ± 0.070	0.670 ± 0.069	0.625 ± 0.029	0.660 ± 0.120	0.660 ± 0.119	0.501 ± 0.159	0.518 ± 0.148	0.611 ± 0.187	0.519 ± 0.040	–	–	–
Framingham	0.616 ± 0.140	0.649 ± 0.099	0.683 ± 0.008	**0.684 ± 0.015**	0.667 ± 0.024	0.674 ± 0.046	0.660 ± 0.046	0.613 ± 0.132	0.668 ± 0.046	0.472 ± 0.125	–	–	–
Blood Transfusion	0.427 ± 0.927	0.511 ± 0.195	0.000 ± 0.000	**0.695 ± 0.128**	0.667 ± 0.067	0.645 ± 0.163	0.682 ± 0.061	0.623 ± 0.155	0.629 ± 0.228	0.542 ± 0.304	–	–	–
Credit-G	0.663 ± 0.127	0.685 ± 0.012	0.700 ± 0.000	**0.717 ± 0.052**	0.687 ± 0.026	0.683 ± 0.133	0.678 ± 0.093	0.700 ± 0.012	0.668 ± 0.050	0.540 ± 0.388	–	–	–
QSAR Biodeg	**0.820 ± 0.096**	0.660 ± 0.018	0.724 ± 0.094	0.771 ± 0.084	0.643 ± 0.068	0.779 ± 0.058	0.485 ± 0.296	0.806 ± 0.059	0.709 ± 0.106	0.423 ± 0.276	–	–	–
Maternal Risk	0.656 ± 0.262	0.341 ± 0.109	0.000 ± 0.000	0.487 ± 0.192	0.436 ± 0.140	0.524 ± 0.417	0.381 ± 0.316	0.626 ± 0.082	0.557 ± 0.057	0.289 ± 0.356	–	–	**0.681**
Website Phishing	**0.761 ± 0.049**	0.441 ± 0.062	0.749 ± 0.071	0.510 ± 0.129	0.749 ± 0.055	0.708 ± 0.079	0.434 ± 0.223	0.683 ± 0.030	0.637 ± 0.081	0.336 ± 0.109	–	–	–

Statistical fidelity

Table 4: Jensen–Shannon div. (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.054 ± 0.054	0.094 ± 0.020	0.200 ± 0.033	0.042 ± 0.002	0.194	0.070 ± 0.006	0.192 ± 0.132	0.265 ± 0.048	0.040 ± 0.004	0.189 ± 0.014	0.049	**0.013**	0.069
Breast Cancer	0.062 ± 0.010	0.143 ± 0.007	0.111 ± 0.022	0.027 ± 0.011	0.294	0.072 ± 0.019	0.568 ± 0.357	0.466 ± 0.024	**0.024 ± 0.014**	0.293 ± 0.032	–	–	–
Liver Disorders	0.068 ± 0.052	0.184 ± 0.038	0.129 ± 0.058	0.028 ± 0.005	**0.028**	0.113 ± 0.031	0.118 ± 0.095	0.450 ± 0.074	0.040 ± 0.004	0.316 ± 0.019	–	–	–
Parkinsons	0.082 ± 0.044	0.175 ± 0.029	0.136 ± 0.013	0.056 ± 0.013	0.227	0.120 ± 0.015	0.143 ± 0.175	0.578 ± 0.075	**0.049 ± 0.004**	0.269 ± 0.016	–	–	–
Obesity	**0.022 ± 0.002**	0.044 ± 0.017	0.090 ± 0.015	0.057 ± 0.011	0.044	0.047 ± 0.075	0.126 ± 0.168	0.158 ± 0.006	0.033 ± 0.011	0.200 ± 0.006	–	–	0.051
Diabetes	0.024 ± 0.016	0.030 ± 0.003	0.035 ± 0.014	0.067 ± 0.007	**0.002 ± 0.001**	0.062 ± 0.041	0.174 ± 0.103	0.094 ± 0.055	0.032 ± 0.020	0.225 ± 0.020	0.061	–	0.031
German Credit	0.207 ± 0.571	0.020 ± 0.008	0.171 ± 0.014	0.008 ± 0.003	**0.007 ± 0.001**	0.019 ± 0.001	0.046 ± 0.025	0.067 ± 0.009	0.023 ± 0.002	0.140 ± 0.009	–	–	–
Bank Marketing	0.016 ± 0.004	0.009 ± 0.005	0.040 ± 0.019	0.015 ± 0.006	**0.004 ± 0.003**	0.040 ± 0.016	0.088 ± 0.110	0.083 ± 0.034	0.040 ± 0.017	0.275 ± 0.032	–	–	–
Census Income	0.050 ± 0.039	0.019 ± 0.005	0.055 ± 0.022	0.049 ± 0.060	**0.007 ± 0.003**	0.082 ± 0.067	0.182 ± 0.344	0.145 ± 0.108	0.085 ± 0.029	0.347 ± 0.009	–	–	–
Melanoma	0.064 ± 0.119	0.118 ± 0.005	0.147 ± 0.076	0.038 ± 0.013	0.044 ± 0.022	0.077 ± 0.079	0.187 ± 0.239	0.331 ± 0.072	**0.033 ± 0.005**	0.127 ± 0.033	–	–	–
PBC	0.054 ± 0.086	0.022 ± 0.018	0.072 ± 0.011	**0.005 ± 0.007**	0.006 ± 0.002	0.021 ± 0.011	0.053 ± 0.012	0.093 ± 0.163	0.014 ± 0.009	0.070 ± 0.011	–	–	–
PBC3	0.045 ± 0.023	0.070 ± 0.005	0.141 ± 0.039	**0.022 ± 0.010**	0.155 ± 0.061	0.051 ± 0.009	0.182 ± 0.169	0.303 ± 0.043	0.031 ± 0.002	0.179 ± 0.015	0.041	–	0.056
Retinopathy	0.126 ± 0.376	0.019 ± 0.023	0.168 ± 0.018	0.009 ± 0.004	**0.006 ± 0.005**	0.025 ± 0.022	0.043 ± 0.044	0.092 ± 0.007	0.020 ± 0.010	0.032 ± 0.006	–	–	–
WHAS500	0.034 ± 0.030	0.075 ± 0.051	0.131 ± 0.032	**0.011 ± 0.001**	0.063 ± 0.029	0.030 ± 0.013	0.129 ± 0.116	0.244 ± 0.020	0.015 ± 0.005	0.129 ± 0.010	–	–	0.066
GBSG2	0.047 ± 0.074	0.062 ± 0.044	0.173 ± 0.026	**0.015 ± 0.006**	0.023 ± 0.008	0.081 ± 0.110	0.131 ± 0.218	0.163 ± 0.058	0.016 ± 0.003	0.258 ± 0.017	0.046	–	–
Colon	0.012 ± 0.011	0.020 ± 0.013	0.104 ± 0.051	**0.003 ± 0.001**	0.005 ± 0.002	0.051 ± 0.064	0.104 ± 0.082	0.087 ± 0.019	0.008 ± 0.004	0.132 ± 0.017	–	–	–
GRACE	0.023 ± 0.008	0.085 ± 0.052	0.120 ± 0.043	**0.008 ± 0.001**	0.013 ± 0.005	0.069 ± 0.042	0.225 ± 0.495	0.269 ± 0.055	0.015 ± 0.012	0.149 ± 0.019	–	–	–
TRACE	0.010 ± 0.012	0.055 ± 0.032	0.111 ± 0.078	0.009 ± 0.001	**0.003 ± 0.002**	0.058 ± 0.041	0.100 ± 0.047	0.180 ± 0.130	0.020 ± 0.009	0.134 ± 0.003	–	–	–
Framingham	0.025 ± 0.005	0.059 ± 0.052	0.095 ± 0.012	0.010 ± 0.003	**0.004 ± 0.002**	0.060 ± 0.020	0.297 ± 0.057	0.226 ± 0.165	0.035 ± 0.011	0.205 ± 0.008	–	–	–
Blood Transfusion	0.123 ± 0.173	0.135 ± 0.061	0.193 ± 0.072	0.037 ± 0.018	**0.029 ± 0.004**	0.173 ± 0.049	0.228 ± 0.092	0.425 ± 0.105	0.043 ± 0.020	0.333 ± 0.025	–	–	–
Credit-G	0.185 ± 0.003	0.020 ± 0.013	0.166 ± 0.034	**0.007 ± 0.001**	0.008 ± 0.002	0.019 ± 0.016	0.067 ± 0.091	0.063 ± 0.006	0.021 ± 0.009	0.143 ± 0.004	–	–	–
QSAR Biodeg	0.071 ± 0.009	0.065 ± 0.014	0.090 ± 0.010	0.094 ± 0.027	0.509 ± 0.001	0.039 ± 0.016	0.272 ± 0.226	0.144 ± 0.022	**0.029 ± 0.005**	0.391 ± 0.011	–	–	–
Maternal Risk	**0.043 ± 0.035**	0.289 ± 0.069	0.264 ± 0.025	0.249 ± 0.020	0.275 ± 0.119	0.235 ± 0.132	0.362 ± 0.175	0.476 ± 0.047	0.140 ± 0.020	0.360 ± 0.014	–	–	0.167
Website Phishing	0.100 ± 0.394	0.224 ± 0.001	0.259 ± 0.003	0.223 ± 0.000	**0.001 ± 0.001**	0.005 ± 0.006	0.060 ± 0.052	0.003 ± 0.005	0.008 ± 0.005	0.028 ± 0.004	–	–	–

Table 5: KS statistic (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.190 ± 0.134	0.201 ± 0.045	0.335 ± 0.035	0.084 ± 0.022	0.262	0.162 ± 0.021	0.356 ± 0.130	0.246 ± 0.036	0.101 ± 0.045	0.313 ± 0.032	0.117	**0.052**	0.130
Breast Cancer	0.206 ± 0.055	0.329 ± 0.017	0.252 ± 0.085	0.103 ± 0.016	0.348	0.191 ± 0.029	0.736 ± 0.296	0.432 ± 0.062	**0.088 ± 0.040**	0.489 ± 0.028	–	–	–
Liver Disorders	0.229 ± 0.141	0.415 ± 0.127	0.265 ± 0.099	0.074 ± 0.014	**0.073**	0.241 ± 0.021	0.278 ± 0.215	0.358 ± 0.098	0.123 ± 0.082	0.479 ± 0.063	–	–	–
Parkinsons	0.184 ± 0.046	0.382 ± 0.051	0.268 ± 0.028	**0.098 ± 0.016**	0.293	0.253 ± 0.069	0.301 ± 0.247	0.454 ± 0.050	0.112 ± 0.017	0.454 ± 0.037	–	–	–
Obesity	0.258 ± 0.011	0.239 ± 0.111	0.235 ± 0.033	0.216 ± 0.024	**0.156**	0.217 ± 0.063	0.354 ± 0.349	0.289 ± 0.023	0.161 ± 0.023	0.305 ± 0.017	–	–	0.193
Diabetes	0.150 ± 0.075	0.121 ± 0.025	0.097 ± 0.020	0.120 ± 0.042	**0.014 ± 0.003**	0.129 ± 0.030	0.262 ± 0.082	0.158 ± 0.058	0.130 ± 0.050	0.400 ± 0.028	0.116	–	0.090
German Credit	0.148 ± 0.144	0.125 ± 0.034	0.381 ± 0.075	0.046 ± 0.022	**0.039 ± 0.002**	0.099 ± 0.049	0.249 ± 0.094	0.143 ± 0.012	0.095 ± 0.041	0.359 ± 0.017	–	–	–
Bank Marketing	0.104 ± 0.024	0.126 ± 0.022	0.171 ± 0.025	0.212 ± 0.048	**0.020 ± 0.007**	0.169 ± 0.144	0.414 ± 0.263	0.285 ± 0.027	0.140 ± 0.018	0.643 ± 0.015	–	–	–
Census Income	0.445 ± 0.094	0.189 ± 0.135	0.227 ± 0.051	0.282 ± 0.273	**0.023 ± 0.004**	0.272 ± 0.021	0.492 ± 0.639	0.211 ± 0.127	0.224 ± 0.167	0.607 ± 0.021	–	–	–
Melanoma	0.182 ± 0.297	0.257 ± 0.006	0.309 ± 0.128	**0.072 ± 0.017**	0.079 ± 0.011	0.156 ± 0.117	0.317 ± 0.285	0.239 ± 0.080	0.098 ± 0.016	0.236 ± 0.016	–	–	–
PBC	0.116 ± 0.054	0.084 ± 0.025	0.193 ± 0.048	0.042 ± 0.045	**0.025 ± 0.014**	0.119 ± 0.058	0.189 ± 0.073	0.147 ± 0.295	0.103 ± 0.044	0.174 ± 0.008	–	–	–
PBC3	0.143 ± 0.067	0.216 ± 0.066	0.243 ± 0.048	**0.073 ± 0.015**	0.205 ± 0.053	0.178 ± 0.041	0.347 ± 0.210	0.302 ± 0.049	0.103 ± 0.027	0.360 ± 0.006	0.112	–	0.134
Retinopathy	0.098 ± 0.077	0.101 ± 0.087	0.370 ± 0.069	0.058 ± 0.027	**0.033 ± 0.016**	0.107 ± 0.050	0.164 ± 0.162	0.138 ± 0.025	0.105 ± 0.053	0.174 ± 0.014	–	–	–
WHAS500	0.115 ± 0.088	0.182 ± 0.072	0.244 ± 0.024	**0.040 ± 0.005**	0.091 ± 0.034	0.092 ± 0.041	0.230 ± 0.154	0.185 ± 0.060	0.065 ± 0.016	0.268 ± 0.017	–	–	0.094
GBSG2	0.158 ± 0.207	0.290 ± 0.109	0.400 ± 0.060	0.132 ± 0.018	0.083 ± 0.029	0.286 ± 0.231	0.397 ± 0.292	0.321 ± 0.052	**0.076 ± 0.012**	0.570 ± 0.040	0.214	–	–
Colon	0.068 ± 0.031	0.100 ± 0.021	0.226 ± 0.063	0.035 ± 0.014	**0.023 ± 0.005**	0.161 ± 0.150	0.254 ± 0.205	0.102 ± 0.028	0.067 ± 0.027	0.320 ± 0.027	–	–	–
GRACE	0.087 ± 0.040	0.190 ± 0.049	0.302 ± 0.057	**0.040 ± 0.013**	0.044 ± 0.015	0.185 ± 0.116	0.363 ± 0.481	0.160 ± 0.028	0.091 ± 0.063	0.241 ± 0.024	–	–	–
TRACE	0.061 ± 0.066	0.142 ± 0.066	0.227 ± 0.111	0.024 ± 0.012	**0.017 ± 0.003**	0.110 ± 0.041	0.230 ± 0.058	0.190 ± 0.231	0.086 ± 0.036	0.285 ± 0.013	–	–	–
Framingham	0.079 ± 0.039	0.174 ± 0.090	0.171 ± 0.035	0.035 ± 0.013	**0.017 ± 0.004**	0.132 ± 0.066	0.436 ± 0.021	0.297 ± 0.357	0.122 ± 0.056	0.316 ± 0.018	–	–	–
Blood Transfusion	0.264 ± 0.336	0.327 ± 0.185	0.396 ± 0.072	0.131 ± 0.038	**0.085 ± 0.021**	0.336 ± 0.008	0.458 ± 0.259	0.280 ± 0.087	0.133 ± 0.064	0.537 ± 0.029	–	–	–
Credit-G	0.110 ± 0.034	0.119 ± 0.027	0.383 ± 0.102	**0.040 ± 0.013**	0.054 ± 0.019	0.113 ± 0.075	0.273 ± 0.324	0.137 ± 0.012	0.091 ± 0.024	0.365 ± 0.012	–	–	–
QSAR Biodeg	0.192 ± 0.065	0.209 ± 0.022	0.204 ± 0.021	0.190 ± 0.031	0.425 ± 0.035	0.131 ± 0.030	0.412 ± 0.171	0.206 ± 0.019	**0.105 ± 0.005**	0.573 ± 0.015	–	–	–
Maternal Risk	0.234 ± 0.041	0.368 ± 0.017	0.352 ± 0.033	0.268 ± 0.040	0.291 ± 0.089	0.291 ± 0.258	0.435 ± 0.263	0.313 ± 0.029	**0.144 ± 0.039**	0.454 ± 0.010	–	–	0.273
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 6: KL divergence (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.396 ± 0.118	0.417 ± 0.141	5.073 ± 2.677	0.218 ± 0.021	0.889	0.739 ± 0.149	0.988 ± 1.095	6.855 ± 1.418	0.261 ± 0.126	0.548 ± 0.087	0.413	**0.152**	0.720
Breast Cancer	0.499 ± 0.214	0.531 ± 0.050	1.804 ± 0.285	0.260 ± 0.236	1.136	0.522 ± 0.361	6.607 ± 11.311	11.996 ± 1.070	**0.173 ± 0.047**	0.801 ± 0.102	–	–	–
Liver Disorders	0.432 ± 0.222	0.889 ± 0.385	3.393 ± 2.232	0.304 ± 0.132	**0.261**	1.205 ± 0.209	0.703 ± 1.668	11.831 ± 0.843	0.367 ± 0.232	0.887 ± 0.088	–	–	–
Parkinsons	1.233 ± 0.905	1.241 ± 0.611	2.603 ± 0.351	**0.430 ± 0.264**	1.399	1.338 ± 0.547	0.973 ± 1.641	15.105 ± 2.132	0.508 ± 0.330	0.766 ± 0.056	–	–	–
Obesity	0.109 ± 0.106	0.122 ± 0.045	1.345 ± 0.151	0.233 ± 0.141	0.124	0.360 ± 0.562	0.456 ± 0.765	4.246 ± 0.378	**0.101 ± 0.038**	0.568 ± 0.006	–	–	0.375
Diabetes	0.104 ± 0.126	0.079 ± 0.006	0.360 ± 0.468	0.175 ± 0.041	**0.006 ± 0.001**	0.380 ± 0.574	0.584 ± 0.449	2.691 ± 0.966	0.093 ± 0.045	0.570 ± 0.056	0.235	–	0.119
German Credit	2.010 ± 2.123	0.058 ± 0.035	3.982 ± 0.208	0.036 ± 0.017	**0.024 ± 0.005**	0.142 ± 0.176	0.180 ± 0.244	1.708 ± 0.256	0.076 ± 0.052	0.368 ± 0.027	–	–	–
Bank Marketing	0.180 ± 0.117	0.028 ± 0.012	0.615 ± 0.457	0.111 ± 0.106	**0.014 ± 0.003**	0.235 ± 0.192	0.300 ± 0.530	2.197 ± 1.182	0.133 ± 0.087	0.819 ± 0.139	–	–	–
Census Income	0.450 ± 0.102	0.062 ± 0.018	0.987 ± 0.673	0.155 ± 0.086	**0.031 ± 0.014**	0.463 ± 0.390	1.749 ± 6.259	3.233 ± 3.534	0.257 ± 0.069	1.137 ± 0.033	–	–	–
Melanoma	0.355 ± 0.307	0.665 ± 0.551	3.288 ± 3.752	0.245 ± 0.161	0.227 ± 0.221	0.812 ± 1.141	1.628 ± 4.267	8.892 ± 1.341	**0.200 ± 0.149**	0.343 ± 0.092	–	–	–
PBC	0.545 ± 1.516	0.059 ± 0.049	2.044 ± 0.410	0.025 ± 0.017	**0.018 ± 0.008**	0.183 ± 0.314	0.157 ± 0.086	1.650 ± 3.313	0.087 ± 0.091	0.175 ± 0.028	–	–	–
PBC3	0.350 ± 0.042	0.323 ± 0.159	3.822 ± 1.163	**0.180 ± 0.084**	0.727 ± 0.314	0.488 ± 0.077	1.163 ± 2.106	7.724 ± 1.264	0.231 ± 0.188	0.502 ± 0.071	0.511	–	0.748
Retinopathy	2.306 ± 9.510	0.054 ± 0.073	4.137 ± 1.474	0.026 ± 0.011	**0.017 ± 0.015**	0.106 ± 0.085	0.178 ± 0.227	2.314 ± 0.104	0.059 ± 0.027	0.092 ± 0.022	–	–	–
WHAS500	0.258 ± 0.113	0.291 ± 0.234	3.193 ± 1.167	**0.091 ± 0.039**	0.233 ± 0.085	0.241 ± 0.108	0.495 ± 0.674	5.838 ± 0.587	0.106 ± 0.104	0.339 ± 0.018	–	–	1.104
GBSG2	0.311 ± 0.195	0.311 ± 0.276	4.253 ± 1.033	**0.093 ± 0.049**	0.115 ± 0.053	0.586 ± 0.792	0.609 ± 1.635	4.183 ± 1.379	0.138 ± 0.071	0.780 ± 0.034	1.170	–	–
Colon	0.108 ± 0.095	0.104 ± 0.067	2.855 ± 0.867	0.027 ± 0.006	**0.016 ± 0.003**	0.273 ± 0.301	0.616 ± 0.901	2.297 ± 1.226	0.031 ± 0.010	0.350 ± 0.047	–	–	–
GRACE	0.289 ± 0.166	0.338 ± 0.247	1.602 ± 1.417	**0.041 ± 0.015**	0.088 ± 0.186	0.862 ± 1.509	2.834 ± 10.957	6.324 ± 1.459	0.085 ± 0.060	0.384 ± 0.060	–	–	–
TRACE	0.093 ± 0.018	0.154 ± 0.095	2.590 ± 4.084	0.024 ± 0.003	**0.009 ± 0.010**	0.304 ± 0.308	0.298 ± 0.127	3.033 ± 1.847	0.059 ± 0.026	0.331 ± 0.009	–	–	–
Framingham	0.312 ± 0.015	0.157 ± 0.139	1.271 ± 0.872	0.044 ± 0.023	**0.011 ± 0.003**	0.407 ± 0.393	1.058 ± 0.287	3.481 ± 6.399	0.114 ± 0.039	0.525 ± 0.027	–	–	–
Blood Transfusion	1.269 ± 2.769	0.563 ± 0.252	3.656 ± 1.451	**0.162 ± 0.038**	0.199 ± 0.234	2.008 ± 0.824	1.265 ± 1.770	11.252 ± 1.432	0.251 ± 0.073	0.906 ± 0.074	–	–	–
Credit-G	3.869 ± 0.059	0.071 ± 0.096	4.069 ± 0.691	**0.026 ± 0.008**	0.027 ± 0.028	0.098 ± 0.075	0.366 ± 0.862	1.565 ± 0.130	0.064 ± 0.038	0.377 ± 0.013	–	–	–
QSAR Biodeg	0.776 ± 0.298	0.263 ± 0.076	2.156 ± 0.388	1.629 ± 1.281	10.681 ± 1.055	0.296 ± 0.175	3.983 ± 4.591	3.813 ± 0.579	**0.180 ± 0.084**	1.125 ± 0.028	–	–	–
Maternal Risk	0.715 ± 0.658	1.003 ± 0.195	5.678 ± 1.994	1.110 ± 0.781	1.200 ± 0.729	1.545 ± 1.718	3.575 ± 4.765	12.353 ± 0.949	**0.412 ± 0.222**	1.023 ± 0.059	–	–	1.232
Website Phishing	0.854 ± 3.562	5.346 ± 0.005	5.952 ± 0.319	5.343 ± 0.002	**0.002 ± 0.002**	0.013 ± 0.017	0.194 ± 0.199	0.009 ± 0.014	0.022 ± 0.011	0.074 ± 0.012	–	–	–

Table 7: Hellinger (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.183 ± 0.099	0.208 ± 0.010	0.419 ± 0.047	0.137 ± 0.008	0.326	0.195 ± 0.025	0.319 ± 0.127	0.376 ± 0.039	0.147 ± 0.013	0.322 ± 0.018	0.147	**0.087**	0.189
Breast Cancer	0.209 ± 0.015	0.313 ± 0.013	0.297 ± 0.028	0.138 ± 0.015	0.490	0.219 ± 0.031	0.668 ± 0.263	0.636 ± 0.019	**0.132 ± 0.033**	0.477 ± 0.027	–	–	–
Liver Disorders	0.219 ± 0.094	0.360 ± 0.045	0.331 ± 0.090	0.147 ± 0.013	**0.147**	0.290 ± 0.047	0.281 ± 0.104	0.633 ± 0.042	0.176 ± 0.011	0.501 ± 0.018	–	–	–
Parkinsons	0.238 ± 0.079	0.357 ± 0.033	0.333 ± 0.018	0.208 ± 0.028	0.427	0.301 ± 0.022	0.320 ± 0.186	0.718 ± 0.051	**0.195 ± 0.010**	0.453 ± 0.017	–	–	–
Obesity	**0.098 ± 0.018**	0.149 ± 0.022	0.245 ± 0.024	0.133 ± 0.011	0.154	0.143 ± 0.126	0.241 ± 0.170	0.277 ± 0.014	0.128 ± 0.028	0.367 ± 0.007	–	–	0.152
Diabetes	0.118 ± 0.043	0.135 ± 0.011	0.142 ± 0.028	0.148 ± 0.005	**0.027 ± 0.007**	0.161 ± 0.056	0.302 ± 0.127	0.208 ± 0.060	0.133 ± 0.038	0.402 ± 0.020	0.160	–	0.116
German Credit	0.309 ± 0.604	0.087 ± 0.010	0.368 ± 0.015	**0.043 ± 0.005**	0.049 ± 0.002	0.082 ± 0.009	0.126 ± 0.043	0.138 ± 0.011	0.112 ± 0.016	0.300 ± 0.010	–	–	–
Bank Marketing	0.102 ± 0.019	0.069 ± 0.021	0.163 ± 0.041	0.065 ± 0.010	**0.046 ± 0.015**	0.112 ± 0.029	0.200 ± 0.145	0.178 ± 0.048	0.140 ± 0.036	0.416 ± 0.020	–	–	–
Census Income	0.182 ± 0.065	0.109 ± 0.020	0.186 ± 0.037	0.111 ± 0.069	**0.062 ± 0.010**	0.194 ± 0.067	0.308 ± 0.328	0.274 ± 0.092	0.213 ± 0.050	0.491 ± 0.007	–	–	–
Melanoma	0.189 ± 0.222	0.249 ± 0.022	0.337 ± 0.113	**0.137 ± 0.028**	0.154 ± 0.042	0.200 ± 0.101	0.328 ± 0.231	0.436 ± 0.036	0.144 ± 0.019	0.269 ± 0.044	–	–	–
PBC	0.164 ± 0.146	0.081 ± 0.010	0.229 ± 0.021	0.045 ± 0.043	**0.039 ± 0.006**	0.098 ± 0.019	0.157 ± 0.078	0.153 ± 0.175	0.090 ± 0.024	0.184 ± 0.017	–	–	–
PBC3	0.165 ± 0.040	0.185 ± 0.018	0.326 ± 0.052	**0.106 ± 0.013**	0.294 ± 0.069	0.170 ± 0.025	0.311 ± 0.156	0.417 ± 0.041	0.142 ± 0.005	0.332 ± 0.016	0.158	–	0.176
Retinopathy	0.212 ± 0.343	0.074 ± 0.022	0.308 ± 0.049	0.050 ± 0.004	**0.040 ± 0.017**	0.105 ± 0.051	0.140 ± 0.067	0.144 ± 0.010	0.094 ± 0.005	0.115 ± 0.013	–	–	–
WHAS500	0.131 ± 0.069	0.172 ± 0.063	0.318 ± 0.037	**0.067 ± 0.005**	0.164 ± 0.039	0.113 ± 0.023	0.233 ± 0.122	0.332 ± 0.045	0.086 ± 0.015	0.273 ± 0.006	–	–	0.161
GBSG2	0.162 ± 0.127	0.167 ± 0.048	0.379 ± 0.021	**0.088 ± 0.009**	0.114 ± 0.021	0.206 ± 0.180	0.254 ± 0.261	0.294 ± 0.049	0.100 ± 0.024	0.396 ± 0.015	0.150	–	–
Colon	0.081 ± 0.046	0.077 ± 0.014	0.274 ± 0.071	**0.038 ± 0.018**	0.038 ± 0.008	0.145 ± 0.107	0.187 ± 0.107	0.162 ± 0.024	0.062 ± 0.017	0.262 ± 0.017	–	–	–
GRACE	0.119 ± 0.038	0.185 ± 0.053	0.282 ± 0.070	**0.062 ± 0.007**	0.085 ± 0.021	0.193 ± 0.069	0.341 ± 0.442	0.383 ± 0.049	0.094 ± 0.032	0.271 ± 0.008	–	–	–
TRACE	0.069 ± 0.065	0.145 ± 0.046	0.277 ± 0.114	0.048 ± 0.011	**0.032 ± 0.011**	0.148 ± 0.045	0.218 ± 0.079	0.275 ± 0.061	0.100 ± 0.021	0.282 ± 0.004	–	–	–
Framingham	0.133 ± 0.003	0.191 ± 0.094	0.206 ± 0.026	0.074 ± 0.014	**0.049 ± 0.012**	0.190 ± 0.036	0.432 ± 0.041	0.382 ± 0.152	0.150 ± 0.025	0.359 ± 0.003	–	–	–
Blood Transfusion	0.289 ± 0.247	0.303 ± 0.106	0.399 ± 0.079	0.166 ± 0.041	**0.149 ± 0.017**	0.364 ± 0.044	0.399 ± 0.075	0.609 ± 0.078	0.178 ± 0.040	0.499 ± 0.029	–	–	–
Credit-G	0.287 ± 0.015	0.081 ± 0.030	0.365 ± 0.039	**0.044 ± 0.002**	0.058 ± 0.015	0.086 ± 0.043	0.165 ± 0.131	0.135 ± 0.002	0.099 ± 0.030	0.304 ± 0.004	–	–	–
QSAR Biodeg	0.208 ± 0.006	0.190 ± 0.009	0.257 ± 0.018	0.200 ± 0.021	0.602 ± 0.008	0.144 ± 0.027	0.401 ± 0.195	0.290 ± 0.020	**0.133 ± 0.013**	0.550 ± 0.008	–	–	–
Maternal Risk	**0.175 ± 0.057**	0.471 ± 0.056	0.470 ± 0.022	0.427 ± 0.009	0.450 ± 0.103	0.397 ± 0.119	0.513 ± 0.162	0.618 ± 0.026	0.304 ± 0.025	0.538 ± 0.010	–	–	0.362
Website Phishing	0.167 ± 0.437	0.247 ± 0.012	0.358 ± 0.005	0.236 ± 0.007	**0.019 ± 0.009**	0.050 ± 0.023	0.184 ± 0.070	0.039 ± 0.033	0.066 ± 0.009	0.121 ± 0.010	–	–	–

Table 8: Pearson corr. preservation (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.927 ± 0.008	0.904 ± 0.030	0.926 ± 0.013	0.930 ± 0.012	0.822	0.912 ± 0.018	0.844 ± 0.042	0.918 ± 0.012	0.908 ± 0.011	0.900 ± 0.005	0.897	**0.939**	0.883
Breast Cancer	0.868 ± 0.033	0.606 ± 0.006	0.799 ± 0.009	**0.948 ± 0.018**	0.773	0.887 ± 0.039	0.566 ± 0.047	0.753 ± 0.028	0.694 ± 0.048	0.607 ± 0.012	–	–	–
Liver Disorders	0.917 ± 0.103	0.744 ± 0.034	0.831 ± 0.125	**0.948 ± 0.005**	0.912	0.862 ± 0.092	0.640 ± 0.144	0.734 ± 0.068	0.814 ± 0.021	0.743 ± 0.034	–	–	–
Parkinsons	0.845 ± 0.085	0.487 ± 0.042	0.653 ± 0.012	**0.936 ± 0.010**	0.739	0.814 ± 0.054	0.481 ± 0.057	0.606 ± 0.043	0.610 ± 0.052	0.486 ± 0.030	–	–	–
Obesity	0.945 ± 0.024	0.871 ± 0.007	0.895 ± 0.014	0.941 ± 0.014	0.847	**0.953 ± 0.088**	0.856 ± 0.053	0.923 ± 0.004	0.892 ± 0.022	0.874 ± 0.010	–	–	0.940
Diabetes	0.928 ± 0.102	0.898 ± 0.033	0.934 ± 0.037	0.916 ± 0.011	**0.986 ± 0.004**	0.968 ± 0.008	0.882 ± 0.039	0.905 ± 0.015	0.918 ± 0.032	0.868 ± 0.013	0.934	–	0.952
German Credit	0.926 ± 0.061	0.896 ± 0.015	0.885 ± 0.052	**0.933 ± 0.020**	0.923 ± 0.057	0.933 ± 0.023	0.879 ± 0.022	0.927 ± 0.029	0.905 ± 0.004	0.897 ± 0.008	–	–	–
Bank Marketing	0.952 ± 0.025	0.943 ± 0.028	0.896 ± 0.062	0.966 ± 0.022	0.824 ± 0.174	**0.969 ± 0.016**	0.922 ± 0.008	0.969 ± 0.031	0.947 ± 0.006	0.940 ± 0.002	–	–	–
Census Income	0.951 ± 0.029	0.953 ± 0.018	0.942 ± 0.004	**0.980 ± 0.005**	0.864 ± 0.038	0.970 ± 0.013	0.914 ± 0.030	0.961 ± 0.014	0.946 ± 0.021	0.941 ± 0.005	–	–	–
Melanoma	**0.910 ± 0.054**	0.832 ± 0.032	0.888 ± 0.035	0.872 ± 0.060	0.889 ± 0.034	0.855 ± 0.119	0.794 ± 0.028	0.848 ± 0.022	0.865 ± 0.109	0.827 ± 0.065	–	–	–
PBC	0.877 ± 0.148	0.859 ± 0.043	0.709 ± 0.220	0.897 ± 0.048	**0.941 ± 0.015**	0.924 ± 0.023	0.862 ± 0.055	0.887 ± 0.034	0.897 ± 0.063	0.861 ± 0.029	–	–	–
PBC3	0.912 ± 0.031	0.864 ± 0.011	0.883 ± 0.051	**0.921 ± 0.017**	0.860 ± 0.027	0.916 ± 0.030	0.857 ± 0.023	0.889 ± 0.007	0.882 ± 0.014	0.863 ± 0.021	0.857	–	0.870
Retinopathy	0.958 ± 0.041	0.781 ± 0.063	0.820 ± 0.177	0.824 ± 0.110	**0.959 ± 0.060**	0.868 ± 0.017	0.784 ± 0.037	0.852 ± 0.077	0.813 ± 0.027	0.779 ± 0.034	–	–	–
WHAS500	0.918 ± 0.025	0.891 ± 0.015	0.903 ± 0.044	0.931 ± 0.010	0.891 ± 0.039	**0.931 ± 0.011**	0.870 ± 0.023	0.917 ± 0.003	0.914 ± 0.010	0.897 ± 0.009	–	–	0.892
GBSG2	0.902 ± 0.088	0.853 ± 0.040	0.871 ± 0.062	**0.920 ± 0.042**	0.820 ± 0.186	0.882 ± 0.116	0.803 ± 0.059	0.884 ± 0.081	0.887 ± 0.048	0.848 ± 0.018	0.886	–	–
Colon	0.957 ± 0.026	0.921 ± 0.013	0.807 ± 0.061	0.942 ± 0.032	**0.961 ± 0.015**	0.915 ± 0.062	0.911 ± 0.004	0.947 ± 0.005	0.931 ± 0.009	0.917 ± 0.013	–	–	–
GRACE	0.951 ± 0.031	0.916 ± 0.036	0.873 ± 0.061	**0.958 ± 0.006**	0.947 ± 0.061	0.942 ± 0.031	0.883 ± 0.074	0.918 ± 0.014	0.950 ± 0.036	0.919 ± 0.041	–	–	–
TRACE	0.965 ± 0.018	0.870 ± 0.003	0.907 ± 0.023	0.930 ± 0.005	**0.982 ± 0.007**	0.931 ± 0.058	0.853 ± 0.021	0.877 ± 0.046	0.924 ± 0.057	0.876 ± 0.009	–	–	–
Framingham	0.969 ± 0.012	0.846 ± 0.031	0.740 ± 0.050	**0.983 ± 0.004**	0.968 ± 0.026	0.964 ± 0.004	0.792 ± 0.016	0.887 ± 0.017	0.883 ± 0.025	0.833 ± 0.006	–	–	–
Blood Transfusion	0.683 ± 0.613	0.548 ± 0.016	0.729 ± 0.060	0.931 ± 0.038	**0.969 ± 0.051**	0.880 ± 0.110	0.576 ± 0.145	0.680 ± 0.163	0.731 ± 0.142	0.549 ± 0.092	–	–	–
Credit-G	**0.959 ± 0.033**	0.892 ± 0.024	0.904 ± 0.027	0.933 ± 0.021	0.905 ± 0.079	0.930 ± 0.033	0.889 ± 0.033	0.930 ± 0.002	0.903 ± 0.016	0.888 ± 0.009	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
QSAR Biodeg	0.791 ± 0.045	0.829 ± 0.002	0.856 ± 0.036	$\mathbf{0.912 \pm 0.013}$	0.822 ± 0.026	0.906 ± 0.010	0.809 ± 0.025	0.873 ± 0.002	0.837 ± 0.014	0.829 ± 0.003	–	–	–
Maternal Risk	0.916 ± 0.059	0.755 ± 0.008	0.832 ± 0.035	$\mathbf{0.929 \pm 0.017}$	0.794 ± 0.155	0.906 ± 0.081	0.751 ± 0.036	0.882 ± 0.039	0.822 ± 0.033	0.750 ± 0.035	–	–	0.883
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 9: JS numeric (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.054 ± 0.054	0.094 ± 0.020	0.200 ± 0.033	0.042 ± 0.002	0.194	0.070 ± 0.006	0.192 ± 0.132	0.265 ± 0.048	0.040 ± 0.004	0.189 ± 0.014	0.049	**0.013**	0.069
Breast Cancer	0.062 ± 0.010	0.143 ± 0.007	0.111 ± 0.022	0.027 ± 0.011	0.294	0.072 ± 0.019	0.568 ± 0.357	0.466 ± 0.024	$\mathbf{0.024 \pm 0.014}$	0.293 ± 0.032	–	–	–
Liver Disorders	0.068 ± 0.052	0.184 ± 0.038	0.129 ± 0.058	0.028 ± 0.005	**0.028**	0.113 ± 0.031	0.118 ± 0.095	0.450 ± 0.074	0.040 ± 0.004	0.316 ± 0.019	–	–	–
Parkinsons	0.082 ± 0.044	0.175 ± 0.029	0.136 ± 0.013	0.056 ± 0.013	0.227	0.120 ± 0.015	0.143 ± 0.175	0.578 ± 0.075	$\mathbf{0.049 \pm 0.004}$	0.269 ± 0.016	–	–	–
Obesity	$\mathbf{0.040 \pm 0.003}$	0.081 ± 0.036	0.136 ± 0.030	0.114 ± 0.022	0.072	0.093 ± 0.144	0.237 ± 0.334	0.313 ± 0.013	0.052 ± 0.014	0.206 ± 0.008	–	–	0.101
Diabetes	0.028 ± 0.022	0.033 ± 0.003	0.030 ± 0.011	0.089 ± 0.009	$\mathbf{0.002 \pm 0.001}$	0.082 ± 0.054	0.216 ± 0.114	0.125 ± 0.072	0.036 ± 0.022	0.248 ± 0.024	0.034	–	0.040
German Credit	0.050 ± 0.076	0.047 ± 0.021	0.202 ± 0.060	0.023 ± 0.009	$\mathbf{0.017 \pm 0.001}$	0.044 ± 0.004	0.111 ± 0.054	0.181 ± 0.026	0.031 ± 0.014	0.160 ± 0.013	–	–	–
Bank Marketing	0.019 ± 0.006	0.010 ± 0.006	0.043 ± 0.030	0.035 ± 0.013	$\mathbf{0.007 \pm 0.007}$	0.027 ± 0.013	0.098 ± 0.163	0.123 ± 0.003	0.025 ± 0.015	0.454 ± 0.076	–	–	–
Census Income	0.052 ± 0.049	0.020 ± 0.008	0.071 ± 0.036	0.115 ± 0.139	$\mathbf{0.010 \pm 0.003}$	0.069 ± 0.034	0.188 ± 0.498	0.139 ± 0.098	0.046 ± 0.023	0.445 ± 0.025	–	–	–
Melanoma	0.064 ± 0.119	0.118 ± 0.005	0.147 ± 0.076	0.038 ± 0.013	0.044 ± 0.022	0.077 ± 0.079	0.187 ± 0.239	0.331 ± 0.072	$\mathbf{0.033 \pm 0.005}$	0.127 ± 0.033	–	–	–
PBC	0.039 ± 0.074	0.026 ± 0.022	0.074 ± 0.014	$\mathbf{0.006 \pm 0.008}$	0.007 ± 0.003	0.025 ± 0.014	0.062 ± 0.012	0.111 ± 0.198	0.016 ± 0.012	0.058 ± 0.005	–	–	–
PBC3	0.042 ± 0.028	0.076 ± 0.006	0.126 ± 0.042	$\mathbf{0.024 \pm 0.011}$	0.166 ± 0.064	0.055 ± 0.010	0.197 ± 0.181	0.330 ± 0.047	0.030 ± 0.003	0.194 ± 0.016	0.043	–	0.061
Retinopathy	0.018 ± 0.007	0.032 ± 0.042	0.269 ± 0.021	0.016 ± 0.007	$\mathbf{0.009 \pm 0.007}$	0.036 ± 0.023	0.065 ± 0.082	0.159 ± 0.013	0.031 ± 0.020	0.054 ± 0.012	–	–	–
WHAS500	0.034 ± 0.030	0.075 ± 0.051	0.131 ± 0.032	$\mathbf{0.011 \pm 0.001}$	0.063 ± 0.029	0.030 ± 0.013	0.129 ± 0.116	0.244 ± 0.020	0.015 ± 0.005	0.129 ± 0.010	–	–	0.066
GBSG2	0.038 ± 0.057	0.098 ± 0.071	0.193 ± 0.039	0.023 ± 0.010	0.035 ± 0.012	0.109 ± 0.127	0.198 ± 0.306	0.260 ± 0.092	$\mathbf{0.022 \pm 0.004}$	0.392 ± 0.027	0.069	–	–
Colon	0.011 ± 0.007	0.026 ± 0.016	0.103 ± 0.054	$\mathbf{0.004 \pm 0.001}$	0.006 ± 0.002	0.056 ± 0.077	0.128 ± 0.102	0.110 ± 0.024	0.009 ± 0.006	0.145 ± 0.019	–	–	–
GRACE	0.023 ± 0.008	0.085 ± 0.052	0.120 ± 0.043	$\mathbf{0.008 \pm 0.001}$	0.013 ± 0.005	0.069 ± 0.042	0.225 ± 0.495	0.269 ± 0.055	0.015 ± 0.012	0.149 ± 0.019	–	–	–
TRACE	0.010 ± 0.012	0.055 ± 0.032	0.111 ± 0.078	0.009 ± 0.001	$\mathbf{0.003 \pm 0.002}$	0.058 ± 0.041	0.100 ± 0.047	0.180 ± 0.130	0.020 ± 0.009	0.134 ± 0.003	–	–	–
Framingham	0.025 ± 0.005	0.059 ± 0.052	0.095 ± 0.012	0.010 ± 0.003	$\mathbf{0.004 \pm 0.002}$	0.060 ± 0.020	0.297 ± 0.057	0.226 ± 0.165	0.035 ± 0.011	0.205 ± 0.008	–	–	–
Blood Transfusion	0.123 ± 0.173	0.135 ± 0.061	0.193 ± 0.072	0.037 ± 0.018	$\mathbf{0.029 \pm 0.004}$	0.173 ± 0.049	0.228 ± 0.092	0.425 ± 0.105	0.043 ± 0.020	0.333 ± 0.025	–	–	–
Credit-G	0.026 ± 0.012	0.046 ± 0.025	0.193 ± 0.072	0.019 ± 0.002	$\mathbf{0.017 \pm 0.002}$	0.043 ± 0.037	0.145 ± 0.226	0.170 ± 0.018	0.029 ± 0.008	0.166 ± 0.014	–	–	–
QSAR Biodeg	0.071 ± 0.009	0.065 ± 0.014	0.090 ± 0.010	0.094 ± 0.027	0.509 ± 0.001	0.039 ± 0.016	0.272 ± 0.226	0.144 ± 0.022	$\mathbf{0.029 \pm 0.005}$	0.391 ± 0.011	–	–	–
Maternal Risk	$\mathbf{0.043 \pm 0.035}$	0.289 ± 0.069	0.264 ± 0.025	0.249 ± 0.020	0.275 ± 0.119	0.235 ± 0.132	0.362 ± 0.175	0.476 ± 0.047	0.140 ± 0.020	0.360 ± 0.014	–	–	0.167
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 10: JS categorical (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	–	–	–	–	–	–	–	–	–	–	–	–	–
Breast Cancer	–	–	–	–	–	–	–	–	–	–	–	–	–
Liver Disorders	–	–	–	–	–	–	–	–	–	–	–	–	–
Parkinsons	–	–	–	–	–	–	–	–	–	–	–	–	–
Obesity	0.004 ± 0.003	0.006 ± 0.003	0.044 ± 0.001	$\mathbf{0.000 \pm 0.000}$	0.015	0.002 ± 0.005	0.015 ± 0.018	0.002 ± 0.003	0.014 ± 0.018	0.195 ± 0.012	–	–	0.002
Diabetes	0.010 ± 0.015	0.020 ± 0.013	0.049 ± 0.025	$\mathbf{0.000 \pm 0.000}$	0.000 ± 0.000	0.003 ± 0.005	0.047 ± 0.104	0.002 ± 0.005	0.022 ± 0.028	0.155 ± 0.009	0.144	–	0.003
German Credit	0.290 ± 0.844	0.005 ± 0.001	0.153 ± 0.023	$\mathbf{0.001 \pm 0.000}$	0.001 ± 0.000	0.005 ± 0.000	0.011 ± 0.010	0.006 ± 0.002	0.019 ± 0.005	0.129 ± 0.007	–	–	–
Bank Marketing	0.014 ± 0.005	0.008 ± 0.004	0.038 ± 0.012	$\mathbf{0.000 \pm 0.000}$	0.003 ± 0.001	0.050 ± 0.027	0.079 ± 0.070	0.052 ± 0.061	0.051 ± 0.040	0.135 ± 0.003	–	–	–
Census Income	0.049 ± 0.032	0.018 ± 0.014	0.043 ± 0.014	$\mathbf{0.000 \pm 0.000}$	0.006 ± 0.003	0.092 ± 0.109	0.177 ± 0.239	0.150 ± 0.195	0.114 ± 0.036	0.274 ± 0.003	–	–	–
Melanoma	–	–	–	–	–	–	–	–	–	–	–	–	–
PBC	0.129 ± 0.460	0.004 ± 0.004	0.058 ± 0.008	0.002 ± 0.004	$\mathbf{0.000 \pm 0.001}$	0.004 ± 0.006	0.012 ± 0.019	0.007 ± 0.016	0.005 ± 0.011	0.129 ± 0.048	–	–	–
PBC3	0.077 ± 0.039	0.003 ± 0.005	0.307 ± 0.035	$\mathbf{0.001 \pm 0.002}$	0.028 ± 0.051	0.008 ± 0.016	0.023 ± 0.037	0.008 ± 0.013	0.050 ± 0.013	0.007 ± 0.007	0.014	–	0.003
Retinopathy	0.270 ± 0.868	0.001 ± 0.001	0.034 ± 0.051	$\mathbf{0.000 \pm 0.000}$	0.001 ± 0.001	0.011 ± 0.025	0.013 ± 0.017	0.001 ± 0.005	0.006 ± 0.006	0.003 ± 0.003	–	–	–
WHAS500	–	–	–	–	–	–	–	–	–	–	–	–	–
GBSG2	0.061 ± 0.202	0.002 ± 0.004	0.139 ± 0.009	$\mathbf{0.001 \pm 0.001}$	0.001 ± 0.001	0.033 ± 0.083	0.018 ± 0.070	0.001 ± 0.001	0.005 ± 0.013	0.035 ± 0.016	0.007	–	–
Colon	0.012 ± 0.022	0.002 ± 0.001	0.108 ± 0.045	0.001 ± 0.001	$\mathbf{0.001 \pm 0.001}$	0.032 ± 0.034	0.018 ± 0.019	0.006 ± 0.006	0.007 ± 0.010	0.087 ± 0.010	–	–	–
GRACE	–	–	–	–	–	–	–	–	–	–	–	–	–
TRACE	–	–	–	–	–	–	–	–	–	–	–	–	–
Framingham	–	–	–	–	–	–	–	–	–	–	–	–	–
Blood Transfusion	–	–	–	–	–	–	–	–	–	–	–	–	–
Credit-G	0.271 ± 0.003	0.006 ± 0.007	0.151 ± 0.015	$\mathbf{0.001 \pm 0.000}$	0.003 ± 0.002	0.006 ± 0.007	0.025 ± 0.044	0.005 ± 0.001	0.017 ± 0.016	0.131 ± 0.001	–	–	–
QSAR Biodeg	–	–	–	–	–	–	–	–	–	–	–	–	–
Maternal Risk	–	–	–	–	–	–	–	–	–	–	–	–	–
Website Phishing	0.100 ± 0.394	0.224 ± 0.001	0.259 ± 0.003	0.223 ± 0.000	$\mathbf{0.001 \pm 0.001}$	0.005 ± 0.006	0.060 ± 0.052	0.003 ± 0.005	0.008 ± 0.005	0.028 ± 0.004	–	–	–

Privacy / disclosure

Table 11: DCR median (↑=less memorization) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.731 ± 0.064	0.869 ± 0.158	0.553 ± 0.234	0.736 ± 0.007	**1.971**	0.703 ± 0.090	1.043 ± 0.312	0.617 ± 0.114	0.715 ± 0.050	1.873 ± 0.111	0.655	0.684	0.611
Breast Cancer	0.550 ± 0.162	1.245 ± 0.120	0.476 ± 0.047	0.555 ± 0.037	**2.769**	0.672 ± 0.083	1.726 ± 0.372	0.433 ± 0.045	0.850 ± 0.176	2.494 ± 0.208	–	–	–
Liver Disorders	0.412 ± 0.158	0.682 ± 0.169	0.240 ± 0.013	0.382 ± 0.048	0.307	0.316 ± 0.036	0.570 ± 0.144	0.379 ± 0.087	0.376 ± 0.124	$\mathbf{1.677 \pm 0.319}$	–	–	–
Parkinsons	0.483 ± 0.222	1.033 ± 0.097	0.510 ± 0.052	0.538 ± 0.063	0.928	0.625 ± 0.124	1.019 ± 0.193	0.457 ± 0.028	0.742 ± 0.102	$\mathbf{1.923 \pm 0.271}$	–	–	–
Obesity	0.277 ± 0.035	0.437 ± 0.042	0.247 ± 0.025	0.403 ± 0.032	0.325	0.342 ± 0.088	0.471 ± 0.041	0.349 ± 0.023	0.426 ± 0.019	$\mathbf{0.706 \pm 0.031}$	–	–	0.299
Diabetes	0.053 ± 0.042	0.082 ± 0.021	0.038 ± 0.009	0.116 ± 0.021	0.056 ± 0.001	0.065 ± 0.008	0.118 ± 0.169	0.068 ± 0.041	0.081 ± 0.021	$\mathbf{0.545 \pm 0.125}$	0.033	–	0.047
German Credit	0.432 ± 0.113	0.456 ± 0.004	0.308 ± 0.040	0.426 ± 0.008	0.392 ± 0.005	0.418 ± 0.023	0.431 ± 0.001	0.402 ± 0.022	0.446 ± 0.024	$\mathbf{0.642 \pm 0.028}$	–	–	–
Bank Marketing	0.160 ± 0.082	0.254 ± 0.020	0.096 ± 0.006	0.269 ± 0.015	0.211 ± 0.002	0.250 ± 0.039	0.331 ± 0.128	0.226 ± 0.033	0.289 ± 0.037	$\mathbf{2.937 \pm 5.289}$	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Census Income	0.087 ± 0.058	0.195 ± 0.012	0.044 ± 0.007	0.254 ± 0.137	0.140 ± 0.000	0.188 ± 0.056	0.247 ± 0.050	0.189 ± 0.077	0.222 ± 0.039	$\mathbf{0.683 \pm 0.097}$	–	–	–
Melanoma	0.364 ± 0.093	0.655 ± 0.209	0.322 ± 0.130	0.404 ± 0.106	0.360 ± 0.028	0.375 ± 0.047	0.748 ± 0.091	0.348 ± 0.097	0.419 ± 0.084	$\mathbf{0.891 \pm 0.076}$	–	–	–
PBC	0.247 ± 0.707	0.201 ± 0.064	0.080 ± 0.065	0.130 ± 0.105	0.080 ± 0.006	0.075 ± 0.031	0.158 ± 0.152	0.242 ± 0.773	0.084 ± 0.007	$\mathbf{0.571 \pm 0.057}$	–	–	–
PBC3	0.568 ± 0.107	0.722 ± 0.037	0.457 ± 0.013	0.602 ± 0.006	1.112 ± 0.474	0.600 ± 0.044	0.809 ± 0.259	0.464 ± 0.025	0.575 ± 0.060	$\mathbf{1.414 \pm 0.109}$	0.524	–	0.514
Retinopathy	0.342 ± 0.315	0.472 ± 0.055	0.111 ± 0.107	0.372 ± 0.134	0.259 ± 0.114	0.300 ± 0.012	0.454 ± 0.134	0.321 ± 0.021	0.373 ± 0.161	$\mathbf{0.542 \pm 0.036}$	–	–	–
WHAS500	0.643 ± 0.102	0.934 ± 0.105	0.480 ± 0.138	0.722 ± 0.026	0.725 ± 0.036	0.678 ± 0.070	1.125 ± 0.123	0.571 ± 0.116	0.702 ± 0.018	$\mathbf{1.822 \pm 0.304}$	–	–	0.619
GBSG2	0.311 ± 0.142	0.450 ± 0.042	0.124 ± 0.012	0.325 ± 0.025	0.251 ± 0.027	0.324 ± 0.187	0.534 ± 0.309	0.261 ± 0.049	0.334 ± 0.046	$\mathbf{1.615 \pm 0.146}$	0.242	–	–
Colon	0.185 ± 0.081	0.368 ± 0.082	0.074 ± 0.006	0.258 ± 0.026	0.193 ± 0.051	0.271 ± 0.292	0.430 ± 0.059	0.142 ± 0.079	0.206 ± 0.107	$\mathbf{1.057 \pm 0.203}$	–	–	–
GRACE	0.228 ± 0.035	0.354 ± 0.236	0.177 ± 0.029	0.233 ± 0.028	0.221 ± 0.019	0.207 ± 0.032	0.676 ± 0.890	0.191 ± 0.019	0.209 ± 0.014	$\mathbf{0.975 \pm 0.106}$	–	–	–
TRACE	0.081 ± 0.036	0.233 ± 0.031	0.069 ± 0.073	0.101 ± 0.001	0.087 ± 0.009	0.072 ± 0.025	0.230 ± 0.356	0.238 ± 0.562	0.085 ± 0.017	$\mathbf{0.768 \pm 0.023}$	–	–	–
Framingham	0.306 ± 0.017	0.467 ± 0.003	0.276 ± 0.016	0.325 ± 0.008	0.308 ± 0.010	0.330 ± 0.051	0.724 ± 0.431	0.466 ± 0.464	0.342 ± 0.036	$\mathbf{1.356 \pm 0.084}$	–	–	–
Blood Transfusion	0.260 ± 0.312	0.542 ± 0.116	0.146 ± 0.025	0.196 ± 0.042	0.153 ± 0.017	0.172 ± 0.069	0.313 ± 0.135	0.244 ± 0.035	0.230 ± 0.058	$\mathbf{2.031 \pm 0.130}$	–	–	–
Credit-G	0.422 ± 0.022	0.446 ± 0.030	0.324 ± 0.066	0.425 ± 0.009	0.389 ± 0.009	0.405 ± 0.033	0.453 ± 0.097	0.395 ± 0.013	0.432 ± 0.007	$\mathbf{0.642 \pm 0.025}$	–	–	–
QSAR Biodeg	0.474 ± 0.102	1.089 ± 0.089	0.389 ± 0.022	0.995 ± 0.356	$\mathbf{5.396 \pm 1.573}$	0.591 ± 0.164	1.033 ± 0.515	0.482 ± 0.014	0.829 ± 0.365	4.355 ± 0.251	–	–	–
Maternal Risk	0.260 ± 0.125	0.653 ± 0.325	0.256 ± 0.083	0.442 ± 0.011	1.012 ± 0.821	0.353 ± 0.079	0.486 ± 0.099	0.307 ± 0.084	0.440 ± 0.081	$\mathbf{1.381 \pm 0.189}$	–	–	0.315
Website Phishing	0.322 ± 0.168	$\mathbf{0.400 \pm 0.000}$	$\mathbf{0.400 \pm 0.000}$	$\mathbf{0.400 \pm 0.000}$	0.283 ± 0.000	0.283 ± 0.000	$\mathbf{0.400 \pm 0.000}$	0.283 ± 0.000	0.322 ± 0.168	$\mathbf{0.400 \pm 0.000}$	–	–	–

Table 12: DCR real–real reference — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.796	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.796	0.796	0.796
Breast Cancer	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.468	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	–	–	–
Liver Disorders	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.342	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	–	–	–
Parkinsons	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.538	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	–	–	–
Obesity	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.284	0.276 ± 0.129	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	–	–	0.284
Diabetes	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071	–	0.071
German Credit	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	–	–	–
Bank Marketing	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	–	–	–
Census Income	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	–	–	–
Melanoma	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	–	–	–
PBC	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	–	–	–
PBC3	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.629	–	0.629
Retinopathy	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	–	–	–
WHAS500	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	–	–	0.741
GBSG2	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.371	–	–
Colon	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	–	–	–
GRACE	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	–	–	–
TRACE	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	–	–	–
Framingham	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	–	–	–
Blood Transfusion	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	–	–	–
Credit-G	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	–	–	–
QSAR Biodeg	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	–	–	–
Maternal Risk	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	–	–	0.356
Website Phishing	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	–	–	–

Table 13: DCR encoded (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	2.153 ± 0.553	2.316 ± 0.299	3.165 ± 0.177	1.999 ± 0.143	2.022	2.025 ± 0.662	2.880 ± 0.566	1.582 ± 0.178	1.918 ± 0.340	$\mathbf{3.810 \pm 0.402}$	1.769	1.669	–
Breast Cancer	3.280 ± 0.411	5.484 ± 0.782	3.199 ± 0.489	2.429 ± 0.233	2.308	3.052 ± 0.345	6.717 ± 3.045	2.951 ± 0.615	3.495 ± 0.493	$\mathbf{12.169 \pm 0.265}$	–	–	–
Liver Disorders	0.641 ± 0.058	0.989 ± 0.150	0.570 ± 0.047	0.560 ± 0.121	0.470	0.676 ± 0.243	0.797 ± 0.059	0.937 ± 0.602	0.594 ± 0.033	$\mathbf{1.627 \pm 0.322}$	–	–	–
Parkinsons	2.372 ± 0.729	3.982 ± 0.120	2.595 ± 0.205	1.978 ± 0.149	1.756	2.676 ± 0.238	3.538 ± 0.665	2.529 ± 0.171	2.767 ± 0.446	$\mathbf{8.250 \pm 1.029}$	–	–	–
Obesity	1.166 ± 0.222	1.909 ± 0.085	1.959 ± 0.111	1.860 ± 0.310	1.273	1.516 ± 0.066	2.083 ± 0.472	1.459 ± 0.129	1.770 ± 0.063	$\mathbf{3.013 \pm 0.145}$	–	–	–
Diabetes	0.498 ± 0.059	0.558 ± 0.027	0.546 ± 0.076	0.660 ± 0.105	–	0.517 ± 0.052	0.822 ± 0.185	0.524 ± 0.106	0.568 ± 0.060	$\mathbf{1.604 \pm 0.012}$	0.748	–	–
German Credit	2.974 ± 0.742	3.107 ± 0.154	3.516 ± 0.132	3.007 ± 0.024	–	2.914 ± 0.062	3.187 ± 0.138	2.880 ± 0.062	3.105 ± 0.140	$\mathbf{4.129 \pm 0.139}$	–	–	–
Bank Marketing	1.652 ± 0.047	1.951 ± 0.121	1.893 ± 0.243	2.006 ± 0.020	–	1.994 ± 0.168	2.382 ± 0.464	1.959 ± 0.405	2.111 ± 0.111	$\mathbf{5.609 \pm 1.075}$	–	–	–
Census Income	1.640 ± 0.098	2.048 ± 0.035	1.781 ± 0.231	2.226 ± 0.404	–	2.049 ± 0.320	2.686 ± 0.876	2.221 ± 0.626	2.212 ± 0.155	$\mathbf{4.370 \pm 0.304}$	–	–	–
Melanoma	0.587 ± 0.084	0.887 ± 0.172	$\mathbf{1.618 \pm 1.285}$	0.546 ± 0.127	0.531 ± 0.022	0.597 ± 0.283	1.072 ± 0.963	0.865 ± 0.360	0.525 ± 0.056	0.904 ± 0.178	–	–	–
PBC	0.630 ± 1.356	0.074 ± 0.048	$\mathbf{0.961 \pm 1.641}$	0.070 ± 0.025	0.072 ± 0.018	0.075 ± 0.041	0.171 ± 0.108	0.574 ± 1.595	0.068 ± 0.032	0.193 ± 0.108	–	–	–
PBC3	2.000 ± 0.207	2.375 ± 0.215	2.786 ± 0.354	2.007 ± 0.154	2.111 ± 0.369	2.127 ± 0.176	2.730 ± 0.463	1.971 ± 0.114	2.029 ± 0.322	$\mathbf{3.829 \pm 0.484}$	2.081	–	–
Retinopathy	0.628 ± 1.706	0.796 ± 0.108	$\mathbf{1.554 \pm 0.348}$	0.529 ± 0.297	0.155 ± 0.098	0.352 ± 0.184	0.820 ± 0.071	0.507 ± 0.241	0.617 ± 0.261	0.852 ± 0.202	–	–	–
WHAS500	2.270 ± 0.345	3.165 ± 0.491	3.354 ± 0.195	2.383 ± 0.168	2.221 ± 0.188	2.260 ± 0.094	3.447 ± 0.792	2.218 ± 0.077	2.387 ± 0.362	$\mathbf{4.183 \pm 0.777}$	–	–	–
GBSG2	0.789 ± 0.403	0.955 ± 0.212	1.540 ± 0.206	0.696 ± 0.144	0.636 ± 0.155	0.868 ± 0.408	1.240 ± 0.741	0.742 ± 0.116	0.692 ± 0.103	$\mathbf{2.470 \pm 0.172}$	0.802	–	–
Colon	0.341 ± 0.356	0.541 ± 0.086	1.519 ± 0.242	0.321 ± 0.021	0.328 ± 0.087	0.513 ± 0.233	1.220 ± 1.010	0.433 ± 0.005	0.341 ± 0.016	$\mathbf{1.772 \pm 0.360}$	–	–	–
GRACE	0.354 ± 0.049	0.446 ± 0.101	0.677 ± 0.548	0.316 ± 0.033	0.319 ± 0.018	0.395 ± 0.162	$\mathbf{1.192 \pm 2.913}$	0.603 ± 0.047	0.321 ± 0.026	0.666 ± 0.091	–	–	–
TRACE	0.089 ± 0.027	0.175 ± 0.129	$\mathbf{0.856 \pm 1.248}$	0.069 ± 0.012	0.074 ± 0.025	0.177 ± 0.106	0.238 ± 0.079	0.737 ± 1.768	0.108 ± 0.013	0.294 ± 0.008	–	–	–
Framingham	0.726 ± 0.014	0.932 ± 0.065	1.008 ± 0.220	0.743 ± 0.007	0.718 ± 0.036	0.780 ± 0.126	1.368 ± 0.122	1.252 ± 1.082	0.786 ± 0.089	$\mathbf{1.567 \pm 0.089}$	–	–	–
Blood Transfusion	0.554 ± 0.418	0.565 ± 0.155	0.533 ± 0.311	0.240 ± 0.049	0.207 ± 0.020	0.353 ± 0.125	0.569 ± 0.578	0.542 ± 0.148	0.276 ± 0.073	$\mathbf{0.975 \pm 0.200}$	–	–	–
Credit-G	3.156 ± 0.053	3.063 ± 0.244	3.468 ± 0.185	2.975 ± 0.201	2.755 ± 0.158	2.866 ± 0.088	3.297 ± 0.324	2.866 ± 0.170	3.038 ± 0.155	$\mathbf{4.188 \pm 0.121}$	–	–	–
QSAR Biodeg	3.690 ± 0.252	4.837 ± 0.670	3.695 ± 0.513	4.228 ± 0.410	$\mathbf{22.772 \pm 4.344}$	3.096 ± 0.058	6.437 ± 2.976	3.072 ± 0.274	3.949 ± 0.307	19.399 ± 7.340	–	–	–
Maternal Risk	0.452 ± 0.309	1.075 ± 0.224	1.418 ± 0.418	0.741 ± 0.130	0.739 ± 0.170	0.767 ± 0.248	1.263 ± 0.936	0.714 ± 0.079	0.651 ± 0.015	$\mathbf{1.582 \pm 0.269}$	–	–	–
Website Phishing	1.520 ± 0.456	1.805 ± 0.840	$\mathbf{2.000 \pm 0.000}$	1.805 ± 0.840	1.414 ± 0.000	1.414 ± 0.000	1.609 ± 0.840	1.414 ± 0.000	1.414 ± 0.000	1.414 ± 0.000	–	–	–

Table 14: NNDR (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.892 ± 0.046	0.918 ± 0.017	$\mathbf{0.983 \pm 0.016}$	0.880 ± 0.024	0.912	0.906 ± 0.089	0.930 ± 0.014	0.909 ± 0.067	0.891 ± 0.037	0.927 ± 0.061	0.875	0.880	–
Breast Cancer	$\mathbf{0.980 \pm 0.004}$	0.958 ± 0.022	0.965 ± 0.003	0.931 ± 0.012	0.929	0.949 ± 0.033	0.877 ± 0.241	0.975 ± 0.010	0.956 ± 0.004	0.972 ± 0.025	–	–	–
Liver Disorders	0.922 ± 0.075	0.877 ± 0.049	0.916 ± 0.077	0.834 ± 0.034	0.867	0.897 ± 0.053	0.860 ± 0.027	$\mathbf{1.000 \pm 0.000}$	0.856 ± 0.105	0.864 ± 0.059	–	–	–
Parkinsons	0.951 ± 0.033	0.956 ± 0.040	0.953 ± 0.019	0.907 ± 0.063	0.828	0.935 ± 0.012	0.944 ± 0.036	$\mathbf{0.992 \pm 0.016}$	0.935 ± 0.010	0.969 ± 0.022	–	–	–
Obesity	0.902 ± 0.023	0.929 ± 0.014	$\mathbf{0.963 \pm 0.010}$	0.925 ± 0.020	0.903	0.910 ± 0.019	0.935 ± 0.023	0.910 ± 0.014	0.923 ± 0.002	0.956 ± 0.013	–	–	–
Diabetes	0.831 ± 0.019	0.834 ± 0.010	0.897 ± 0.027	0.847 ± 0.012	–	0.846 ± 0.026	0.871 ± 0.034	0.848 ± 0.056	0.831 ± 0.013	$\mathbf{0.905 \pm 0.021}$	0.863	–	–
German Credit	0.953 ± 0.003	0.960 ± 0.001	$\mathbf{0.986 \pm 0.006}$	0.957 ± 0.017	–	0.951 ± 0.015	0.956 ± 0.007	0.949 ± 0.010	0.953 ± 0.017	0.962 ± 0.005	–	–	–
Bank Marketing	0.927 ± 0.008	0.938 ± 0.008	$\mathbf{0.966 \pm 0.010}$	0.945 ± 0.006	–	0.943 ± 0.003	0.955 ± 0.010	0.949 ± 0.018	0.945 ± 0.008	0.892 ± 0.164	–	–	–
Census Income	0.931 ± 0.008	0.944 ± 0.007	$\mathbf{0.970 \pm 0.009}$	0.948 ± 0.005	–	0.947 ± 0.005	0.962 ± 0.010	0.966 ± 0.025	0.952 ± 0.004	0.957 ± 0.028	–	–	–
Melanoma	0.857 ± 0.153	0.824 ± 0.076	0.922 ± 0.056	0.809 ± 0.051	0.807 ± 0.156	0.830 ± 0.053	0.832 ± 0.049	$\mathbf{1.000 \pm 0.000}$	0.805 ± 0.071	0.759 ± 0.029	–	–	–
PBC	0.712 ± 0.379	0.595 ± 0.172	$\mathbf{0.886 \pm 0.319}$	0.516 ± 0.074	0.498 ± 0.048	0.542 ± 0.145	0.638 ± 0.298	0.817 ± 0.409	0.538 ± 0.026	0.493 ± 0.096	–	–	–
PBC3	0.917 ± 0.035	0.916 ± 0.022	$\mathbf{0.970 \pm 0.008}$	0.899 ± 0.015	0.902 ± 0.006	0.911 ± 0.033	0.918 ± 0.024	0.940 ± 0.030	0.916 ± 0.034	0.932 ± 0.029	0.920	–	–
Retinopathy	0.725 ± 0.617	0.796 ± 0.041	0.979 ± 0.049	0.727 ± 0.120	0.441 ± 0.202	0.688 ± 0.059	0.810 ± 0.093	$\mathbf{1.000 \pm 0.000}$	0.680 ± 0.274	0.772 ± 0.036	–	–	–
WHAS500	0.933 ± 0.027	0.932 ± 0.027	$\mathbf{0.980 \pm 0.015}$	0.919 ± 0.010	0.895 ± 0.093	0.897 ± 0.072	0.926 ± 0.026	0.929 ± 0.013	0.907 ± 0.056	0.910 ± 0.073	–	–	–
GBSG2	0.871 ± 0.085	0.880 ± 0.013	0.971 ± 0.005	0.867 ± 0.044	0.853 ± 0.053	0.887 ± 0.038	0.888 ± 0.088	$\mathbf{0.973 \pm 0.055}$	0.850 ± 0.019	0.847 ± 0.087	0.901	–	–
Colon	0.734 ± 0.207	0.759 ± 0.044	0.985 ± 0.002	0.710 ± 0.098	0.688 ± 0.148	0.791 ± 0.091	0.841 ± 0.085	$\mathbf{1.000 \pm 0.000}$	0.703 ± 0.028	0.884 ± 0.055	–	–	–
GRACE	0.814 ± 0.058	0.806 ± 0.052	0.878 ± 0.104	0.762 ± 0.087	0.798 ± 0.026	0.824 ± 0.102	0.815 ± 0.068	$\mathbf{1.000 \pm 0.000}$	0.787 ± 0.066	0.789 ± 0.083	–	–	–
TRACE	0.720 ± 0.084	0.641 ± 0.101	0.894 ± 0.150	0.586 ± 0.060	0.616 ± 0.082	0.786 ± 0.114	0.727 ± 0.024	$\mathbf{0.980 \pm 0.086}$	0.734 ± 0.056	0.687 ± 0.056	–	–	–
Framingham	0.895 ± 0.013	0.892 ± 0.004	$\mathbf{0.924 \pm 0.007}$	0.888 ± 0.018	0.888 ± 0.012	0.898 ± 0.021	0.898 ± 0.028	0.919 ± 0.037	0.895 ± 0.004	0.883 ± 0.014	–	–	–
Blood Transfusion	0.946 ± 0.081	0.859 ± 0.114	0.891 ± 0.049	0.758 ± 0.150	0.795 ± 0.040	0.870 ± 0.109	0.846 ± 0.037	$\mathbf{1.000 \pm 0.000}$	0.823 ± 0.019	0.805 ± 0.133	–	–	–
Credit-G	0.964 ± 0.005	0.955 ± 0.013	$\mathbf{0.989 \pm 0.013}$	0.954 ± 0.021	0.948 ± 0.004	0.948 ± 0.020	0.960 ± 0.010	0.948 ± 0.008	0.954 ± 0.009	0.963 ± 0.021	–	–	–
QSAR Biodeg	0.970 ± 0.009	0.959 ± 0.021	$\mathbf{0.978 \pm 0.020}$	0.959 ± 0.019	0.972 ± 0.034	0.949 ± 0.009	0.970 ± 0.028	0.949 ± 0.006	0.961 ± 0.011	0.915 ± 0.284	–	–	–
Maternal Risk	0.964 ± 0.146	0.886 ± 0.059	0.975 ± 0.002	0.889 ± 0.035	0.876 ± 0.015	0.876 ± 0.082	0.889 ± 0.054	$\mathbf{1.000 \pm 0.000}$	0.861 ± 0.043	0.898 ± 0.031	–	–	–
Website Phishing	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	0.902 ± 0.420	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	–	–	–

Table 15: NNAA (0.5 ideal) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.604 ± 0.174	0.738 ± 0.052	0.967 ± 0.038	0.508 ± 0.131	0.655	0.600 ± 0.050	0.875 ± 0.151	0.629 ± 0.086	0.536 ± 0.023	0.927 ± 0.014	0.582	0.452	–
Breast Cancer	0.975 ± 0.000	0.958 ± 0.029	0.944 ± 0.017	0.639 ± 0.058	0.713	0.882 ± 0.055	0.996 ± 0.007	0.932 ± 0.030	0.753 ± 0.032	0.951 ± 0.003	–	–	–
Liver Disorders	0.361 ± 0.076	0.850 ± 0.037	0.673 ± 0.145	0.505 ± 0.080	0.458	0.711 ± 0.060	0.772 ± 0.160	0.905 ± 0.077	0.562 ± 0.070	0.891 ± 0.043	–	–	–
Parkinsons	0.903 ± 0.072	0.953 ± 0.017	0.934 ± 0.059	0.637 ± 0.053	0.638	0.907 ± 0.029	0.886 ± 0.122	0.936 ± 0.036	0.740 ± 0.072	0.893 ± 0.056	–	–	–
Obesity	0.591 ± 0.013	0.844 ± 0.027	0.935 ± 0.010	0.864 ± 0.088	0.726	0.775 ± 0.079	0.900 ± 0.109	0.761 ± 0.014	0.819 ± 0.062	0.950 ± 0.009	–	–	–
Diabetes	0.585 ± 0.036	0.628 ± 0.015	0.682 ± 0.042	0.691 ± 0.060	–	0.630 ± 0.065	0.828 ± 0.124	0.624 ± 0.155	0.636 ± 0.059	0.920 ± 0.013	0.684	–	–
German Credit	0.790 ± 0.393	0.703 ± 0.112	0.948 ± 0.003	0.614 ± 0.019	–	0.578 ± 0.048	0.775 ± 0.076	0.550 ± 0.011	0.686 ± 0.039	0.910 ± 0.014	–	–	–
Bank Marketing	0.612 ± 0.041	0.703 ± 0.034	0.806 ± 0.097	0.750 ± 0.033	–	0.775 ± 0.084	0.903 ± 0.087	0.784 ± 0.220	0.791 ± 0.040	0.995 ± 0.005	–	–	–
Census Income	0.663 ± 0.068	0.734 ± 0.034	0.789 ± 0.078	0.840 ± 0.163	–	0.828 ± 0.179	0.944 ± 0.119	0.908 ± 0.165	0.868 ± 0.029	0.990 ± 0.003	–	–	–
Melanoma	0.536 ± 0.098	0.744 ± 0.026	0.787 ± 0.169	0.576 ± 0.077	0.467 ± 0.083	0.641 ± 0.203	0.797 ± 0.225	0.884 ± 0.062	0.534 ± 0.160	0.763 ± 0.024	–	–	–
PBC	0.679 ± 0.631	0.566 ± 0.058	0.699 ± 0.045	0.500 ± 0.065	0.477 ± 0.076	0.562 ± 0.055	0.673 ± 0.064	0.802 ± 0.468	0.533 ± 0.055	0.701 ± 0.036	–	–	–
PBC3	0.556 ± 0.102	0.740 ± 0.084	0.906 ± 0.038	0.553 ± 0.069	0.596 ± 0.041	0.643 ± 0.049	0.880 ± 0.203	0.719 ± 0.035	0.624 ± 0.062	0.912 ± 0.005	0.670	–	–
Retinopathy	0.571 ± 0.897	0.797 ± 0.057	0.734 ± 0.097	0.635 ± 0.104	0.315 ± 0.035	0.550 ± 0.178	0.799 ± 0.084	0.691 ± 0.076	0.661 ± 0.089	0.811 ± 0.057	–	–	–
WHAS500	0.674 ± 0.034	0.808 ± 0.120	0.951 ± 0.027	0.533 ± 0.018	0.497 ± 0.111	0.586 ± 0.020	0.898 ± 0.169	0.654 ± 0.125	0.594 ± 0.080	0.912 ± 0.017	–	–	–
GBSG2	0.634 ± 0.319	0.778 ± 0.066	0.923 ± 0.033	0.579 ± 0.015	0.502 ± 0.001	0.750 ± 0.244	0.852 ± 0.164	0.749 ± 0.073	0.593 ± 0.071	0.959 ± 0.009	0.665	–	–
Colon	0.449 ± 0.206	0.649 ± 0.057	0.708 ± 0.102	0.491 ± 0.023	0.411 ± 0.020	0.699 ± 0.188	0.872 ± 0.220	0.762 ± 0.054	0.568 ± 0.083	0.875 ± 0.030	–	–	–
GRACE	0.504 ± 0.022	0.720 ± 0.079	0.797 ± 0.077	0.539 ± 0.035	0.507 ± 0.026	0.672 ± 0.141	0.821 ± 0.399	0.919 ± 0.027	0.536 ± 0.031	0.831 ± 0.029	–	–	–
TRACE	0.536 ± 0.063	0.696 ± 0.065	0.820 ± 0.145	0.537 ± 0.012	0.507 ± 0.009	0.761 ± 0.060	0.830 ± 0.033	0.946 ± 0.048	0.645 ± 0.028	0.830 ± 0.003	–	–	–
Framingham	0.556 ± 0.039	0.740 ± 0.063	0.817 ± 0.050	0.537 ± 0.032	0.495 ± 0.011	0.644 ± 0.165	0.936 ± 0.015	0.839 ± 0.402	0.636 ± 0.043	0.924 ± 0.013	–	–	–
Blood Transfusion	0.862 ± 0.269	0.900 ± 0.081	0.885 ± 0.065	0.627 ± 0.056	0.541 ± 0.066	0.814 ± 0.057	0.912 ± 0.133	0.975 ± 0.007	0.718 ± 0.076	0.956 ± 0.009	–	–	–
Credit-G	0.911 ± 0.015	0.718 ± 0.123	0.939 ± 0.024	0.612 ± 0.021	0.477 ± 0.014	0.594 ± 0.060	0.828 ± 0.226	0.540 ± 0.036	0.691 ± 0.028	0.916 ± 0.010	–	–	–
QSAR Biodeg	0.980 ± 0.015	0.946 ± 0.024	0.989 ± 0.009	0.967 ± 0.012	0.997 ± 0.004	0.890 ± 0.067	0.997 ± 0.002	0.897 ± 0.022	0.893 ± 0.024	0.995 ± 0.006	–	–	–
Maternal Risk	0.668 ± 0.083	0.882 ± 0.023	0.949 ± 0.012	0.807 ± 0.067	0.829 ± 0.058	0.860 ± 0.144	0.929 ± 0.089	0.890 ± 0.007	0.759 ± 0.016	0.911 ± 0.033	–	–	–
Website Phishing	0.368 ± 0.919	0.706 ± 0.071	0.875 ± 0.010	0.687 ± 0.053	0.022 ± 0.012	0.140 ± 0.047	0.520 ± 0.162	0.138 ± 0.026	0.174 ± 0.046	0.409 ± 0.049	–	–	–

Table 16: MIA AUROC (0.5 safe) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.499 ± 0.105	0.470 ± 0.077	0.477 ± 0.120	0.493 ± 0.061	0.486	0.515 ± 0.148	0.491 ± 0.105	0.458 ± 0.046	0.481 ± 0.049	0.492 ± 0.061	0.421	0.444	–
Breast Cancer	0.516 ± 0.100	0.500 ± 0.056	0.497 ± 0.059	0.497 ± 0.070	0.491	0.511 ± 0.140	0.521 ± 0.132	0.499 ± 0.101	0.496 ± 0.095	0.498 ± 0.064	–	–	–
Liver Disorders	0.632 ± 0.045	0.511 ± 0.085	0.500 ± 0.052	0.498 ± 0.078	0.503	0.525 ± 0.087	0.494 ± 0.049	0.514 ± 0.055	0.500 ± 0.063	0.508 ± 0.130	–	–	–
Parkinsons	0.516 ± 0.214	0.501 ± 0.058	0.483 ± 0.110	0.486 ± 0.071	0.510	0.499 ± 0.039	0.487 ± 0.073	0.536 ± 0.118	0.502 ± 0.076	0.536 ± 0.037	–	–	–
Obesity	0.559 ± 0.026	0.494 ± 0.001	0.503 ± 0.007	0.488 ± 0.036	0.527	0.515 ± 0.067	0.492 ± 0.051	0.512 ± 0.024	0.498 ± 0.035	0.499 ± 0.019	–	–	–
Diabetes	0.504 ± 0.016	0.508 ± 0.027	0.504 ± 0.014	0.497 ± 0.022	–	0.497 ± 0.010	0.498 ± 0.006	0.500 ± 0.012	0.490 ± 0.017	0.501 ± 0.022	0.500	–	–
German Credit	0.513 ± 0.056	0.520 ± 0.038	0.506 ± 0.061	0.522 ± 0.031	–	0.530 ± 0.026	0.497 ± 0.021	0.552 ± 0.028	0.511 ± 0.041	0.524 ± 0.044	–	–	–
Bank Marketing	0.505 ± 0.021	0.503 ± 0.034	0.506 ± 0.022	0.504 ± 0.026	–	0.498 ± 0.006	0.502 ± 0.008	0.501 ± 0.006	0.503 ± 0.024	0.498 ± 0.017	–	–	–
Census Income	0.503 ± 0.015	0.493 ± 0.015	0.501 ± 0.014	0.507 ± 0.020	–	0.507 ± 0.020	0.497 ± 0.013	0.507 ± 0.010	0.496 ± 0.028	0.500 ± 0.021	–	–	–
Melanoma	0.647 ± 0.090	0.572 ± 0.060	0.550 ± 0.076	0.526 ± 0.041	0.588 ± 0.064	0.538 ± 0.059	0.513 ± 0.057	0.563 ± 0.117	0.561 ± 0.084	0.486 ± 0.157	–	–	–
PBC	0.571 ± 0.112	0.510 ± 0.078	0.539 ± 0.110	0.529 ± 0.036	0.558 ± 0.020	0.522 ± 0.028	0.511 ± 0.101	0.525 ± 0.083	0.518 ± 0.061	0.508 ± 0.061	–	–	–
PBC3	0.552 ± 0.054	0.502 ± 0.094	0.490 ± 0.052	0.495 ± 0.022	0.602 ± 0.134	0.555 ± 0.113	0.496 ± 0.065	0.533 ± 0.082	0.507 ± 0.094	0.501 ± 0.106	0.520	–	–
Retinopathy	0.601 ± 0.171	0.502 ± 0.016	0.497 ± 0.047	0.509 ± 0.075	0.533 ± 0.018	0.500 ± 0.056	0.491 ± 0.088	0.535 ± 0.078	0.496 ± 0.079	0.499 ± 0.022	–	–	–
WHAS500	0.516 ± 0.088	0.523 ± 0.057	0.497 ± 0.051	0.538 ± 0.058	0.579 ± 0.068	0.519 ± 0.034	0.507 ± 0.038	0.525 ± 0.030	0.499 ± 0.086	0.484 ± 0.057	–	–	–
GBSG2	0.527 ± 0.083	0.482 ± 0.045	0.484 ± 0.079	0.499 ± 0.067	0.512 ± 0.105	0.501 ± 0.067	0.497 ± 0.078	0.495 ± 0.094	0.481 ± 0.064	0.481 ± 0.053	0.510	–	–
Colon	0.520 ± 0.058	0.491 ± 0.023	0.485 ± 0.030	0.466 ± 0.015	0.508 ± 0.046	0.489 ± 0.075	0.482 ± 0.039	0.478 ± 0.028	0.468 ± 0.011	0.516 ± 0.076	–	–	–
GRACE	0.571 ± 0.026	0.476 ± 0.011	0.497 ± 0.051	0.485 ± 0.044	0.524 ± 0.011	0.495 ± 0.064	0.514 ± 0.069	0.502 ± 0.057	0.489 ± 0.012	0.493 ± 0.006	–	–	–
TRACE	0.578 ± 0.006	0.496 ± 0.039	0.498 ± 0.005	0.503 ± 0.060	0.511 ± 0.046	0.498 ± 0.059	0.497 ± 0.010	0.499 ± 0.006	0.508 ± 0.028	0.497 ± 0.042	–	–	–
Framingham	0.512 ± 0.025	0.509 ± 0.022	0.508 ± 0.008	0.507 ± 0.026	0.517 ± 0.038	0.509 ± 0.014	0.505 ± 0.030	0.508 ± 0.024	0.509 ± 0.051	0.507 ± 0.057	–	–	–
Blood Transfusion	0.533 ± 0.094	0.511 ± 0.014	0.507 ± 0.058	0.493 ± 0.217	0.537 ± 0.023	0.505 ± 0.006	0.517 ± 0.058	0.512 ± 0.104	0.526 ± 0.041	0.499 ± 0.102	–	–	–
Credit-G	0.516 ± 0.025	0.499 ± 0.055	0.505 ± 0.098	0.491 ± 0.122	0.542 ± 0.044	0.494 ± 0.050	0.488 ± 0.038	0.533 ± 0.059	0.500 ± 0.067	0.523 ± 0.061	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
QSAR Biodeg	0.503 ± 0.026	0.490 ± 0.058	0.497 ± 0.034	0.514 ± 0.033	0.498 ± 0.051	0.511 ± 0.015	0.485 ± 0.048	0.507 ± 0.021	0.497 ± 0.051	0.484 ± 0.025	–	–	–
Maternal Risk	0.647 ± 0.020	0.508 ± 0.122	0.487 ± 0.010	0.470 ± 0.015	0.511 ± 0.119	0.501 ± 0.142	0.493 ± 0.096	0.502 ± 0.088	0.506 ± 0.112	0.500 ± 0.046	–	–	–
Website Phishing	0.530 ± 0.028	0.502 ± 0.063	0.530 ± 0.017	0.499 ± 0.051	0.708 ± 0.046	0.537 ± 0.073	0.498 ± 0.112	0.513 ± 0.041	0.520 ± 0.030	0.507 ± 0.019	–	–	–

Table 17: Duplicate rate (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000**	–
Breast Cancer	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Liver Disorders	0.307 ± 0.675	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Parkinsons	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Obesity	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Diabetes	**0.000 ± 0.000**	0.000 ± 0.000	0.001 ± 0.001	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	–
German Credit	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Bank Marketing	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Census Income	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Melanoma	0.006 ± 0.026	**0.000 ± 0.000**	0.002 ± 0.009	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
PBC	**0.000 ± 0.000**	0.011 ± 0.015	**0.000 ± 0.000**	**0.000 ± 0.000**	0.020 ± 0.000	**0.000 ± 0.000**	0.003 ± 0.006	0.017 ± 0.075	0.004 ± 0.010	**0.000 ± 0.000**	–	–	–
PBC3	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	–
Retinopathy	0.263 ± 0.584	0.011 ± 0.021	0.327 ± 0.140	0.056 ± 0.027	0.232 ± 0.020	0.119 ± 0.110	0.024 ± 0.022	0.101 ± 0.047	0.056 ± 0.055	**0.008 ± 0.010**	–	–	–
WHAS500	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
GBSG2	0.001 ± 0.003	**0.000 ± 0.000**	0.001 ± 0.003	**0.000 ± 0.000**	0.001 ± 0.003	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	–
Colon	**0.000 ± 0.000**	0.036 ± 0.020	0.330 ± 0.053	0.081 ± 0.019	0.087 ± 0.017	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
GRACE	0.052 ± 0.115	0.000 ± 0.002	0.005 ± 0.010	0.002 ± 0.006	0.006 ± 0.005	0.003 ± 0.003	0.001 ± 0.003	0.000 ± 0.002	0.003 ± 0.005	**0.000 ± 0.000**	–	–	–
TRACE	**0.000 ± 0.000**	0.000 ± 0.001	0.001 ± 0.002	0.001 ± 0.002	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Framingham	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Blood Transfusion	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.033 ± 0.042	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Credit-G	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
QSAR Biodeg	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Maternal Risk	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Website Phishing	0.402 ± 0.866	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.796 ± 0.039	0.343 ± 0.046	0.157 ± 0.079	0.319 ± 0.044	0.298 ± 0.129	0.055 ± 0.005	–	–	–

Diversity / coverage (PRDC)

Table 18: Precision (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.965 ± 0.074	0.812 ± 0.120	**1.000 ± 0.000**	0.943 ± 0.043	0.477	0.967 ± 0.021	0.619 ± 0.360	0.996 ± 0.018	0.940 ± 0.032	0.457 ± 0.146	0.950	0.987	–
Breast Cancer	0.575 ± 0.156	0.004 ± 0.016	0.221 ± 0.149	**0.853 ± 0.090**	0.433	0.350 ± 0.035	0.001 ± 0.006	0.541 ± 0.033	0.035 ± 0.068	0.054 ± 0.121	–	–	–
Liver Disorders	0.995 ± 0.010	0.571 ± 0.315	**0.999 ± 0.005**	0.981 ± 0.005	0.978	0.982 ± 0.024	0.704 ± 0.095	0.998 ± 0.005	0.937 ± 0.050	0.594 ± 0.172	–	–	–
Parkinsons	**0.872 ± 0.126**	0.004 ± 0.018	0.553 ± 0.233	0.848 ± 0.112	0.814	0.372 ± 0.115	0.013 ± 0.016	0.842 ± 0.072	0.158 ± 0.113	0.107 ± 0.096	–	–	–
Obesity	**0.937 ± 0.007**	0.484 ± 0.015	0.874 ± 0.047	0.587 ± 0.042	0.688	0.684 ± 0.331	0.374 ± 0.191	0.702 ± 0.062	0.526 ± 0.095	0.098 ± 0.120	–	–	–
Diabetes	0.982 ± 0.011	0.940 ± 0.035	**0.995 ± 0.005**	0.874 ± 0.099	–	0.965 ± 0.032	0.757 ± 0.675	0.917 ± 0.214	0.916 ± 0.041	0.514 ± 0.068	0.977	–	–
German Credit	0.924 ± 0.199	0.734 ± 0.078	**1.000 ± 0.002**	0.822 ± 0.036	–	0.888 ± 0.016	0.700 ± 0.094	0.926 ± 0.022	0.671 ± 0.019	0.150 ± 0.053	–	–	–
Bank Marketing	0.971 ± 0.023	0.720 ± 0.082	**0.990 ± 0.009**	0.664 ± 0.165	–	0.743 ± 0.089	0.392 ± 0.130	0.761 ± 0.254	0.636 ± 0.049	0.323 ± 0.658	–	–	–
Census Income	0.967 ± 0.025	0.543 ± 0.086	**0.983 ± 0.012**	0.405 ± 0.417	–	0.480 ± 0.355	0.186 ± 0.298	0.377 ± 0.516	0.321 ± 0.141	0.017 ± 0.054	–	–	–
Melanoma	0.990 ± 0.009	0.807 ± 0.187	0.990 ± 0.009	0.961 ± 0.032	0.909 ± 0.055	**0.994 ± 0.026**	0.789 ± 0.091	0.986 ± 0.038	0.953 ± 0.068	0.831 ± 0.057	–	–	–
PBC	0.771 ± 0.755	0.912 ± 0.085	**1.000 ± 0.000**	0.981 ± 0.032	0.997 ± 0.006	0.991 ± 0.012	0.914 ± 0.168	0.888 ± 0.467	0.993 ± 0.021	0.841 ± 0.066	–	–	–
PBC3	0.982 ± 0.036	0.659 ± 0.163	**1.000 ± 0.000**	0.915 ± 0.014	0.568 ± 0.096	0.944 ± 0.094	0.456 ± 0.253	0.995 ± 0.014	0.906 ± 0.037	0.225 ± 0.181	0.953	–	–
Retinopathy	0.704 ± 0.774	0.476 ± 0.081	**0.994 ± 0.013**	0.655 ± 0.106	0.977 ± 0.035	0.815 ± 0.167	0.515 ± 0.182	0.737 ± 0.030	0.639 ± 0.156	0.442 ± 0.086	–	–	–
WHAS500	0.987 ± 0.021	0.596 ± 0.268	0.997 ± 0.007	0.954 ± 0.024	0.785 ± 0.058	0.960 ± 0.031	0.341 ± 0.392	**1.000 ± 0.000**	0.938 ± 0.054	0.725 ± 0.314	–	–	–
GBSG2	0.892 ± 0.368	0.648 ± 0.246	0.927 ± 0.172	0.917 ± 0.044	**0.952 ± 0.068**	0.937 ± 0.053	0.648 ± 0.300	0.916 ± 0.051	0.871 ± 0.072	0.540 ± 0.203	0.863	–	–
Colon	0.994 ± 0.000	0.903 ± 0.090	**0.996 ± 0.013**	0.934 ± 0.016	0.982 ± 0.013	0.976 ± 0.022	0.712 ± 0.280	0.961 ± 0.013	0.928 ± 0.083	0.593 ± 0.073	–	–	–
GRACE	0.994 ± 0.016	0.929 ± 0.038	0.997 ± 0.005	0.980 ± 0.018	0.975 ± 0.016	0.989 ± 0.025	0.560 ± 1.175	**1.000 ± 0.002**	0.988 ± 0.010	0.665 ± 0.016	–	–	–
TRACE	**0.999 ± 0.002**	0.834 ± 0.080	0.980 ± 0.024	0.970 ± 0.012	0.989 ± 0.003	0.995 ± 0.001	0.755 ± 0.208	0.899 ± 0.405	0.982 ± 0.004	0.641 ± 0.036	–	–	–
Framingham	0.980 ± 0.013	0.646 ± 0.137	**0.984 ± 0.013**	0.955 ± 0.001	0.970 ± 0.012	0.938 ± 0.080	0.301 ± 0.513	0.652 ± 1.006	0.904 ± 0.071	0.272 ± 0.078	–	–	–
Blood Transfusion	0.688 ± 0.618	0.345 ± 0.101	0.771 ± 0.088	0.947 ± 0.052	**0.987 ± 0.009**	0.829 ± 0.126	0.553 ± 0.401	0.581 ± 0.036	0.658 ± 0.207	0.347 ± 0.024	–	–	–
Credit-G	0.791 ± 0.094	0.697 ± 0.081	**1.000 ± 0.000**	0.822 ± 0.009	0.966 ± 0.010	0.912 ± 0.048	0.673 ± 0.327	0.933 ± 0.027	0.734 ± 0.032	0.122 ± 0.031	–	–	–
QSAR Biodeg	0.149 ± 0.037	0.052 ± 0.005	**0.419 ± 0.150**	0.093 ± 0.070	0.175 ± 0.154	0.251 ± 0.076	0.006 ± 0.010	0.348 ± 0.041	0.073 ± 0.031	0.040 ± 0.014	–	–	–
Maternal Risk	**0.992 ± 0.018**	0.620 ± 0.360	0.934 ± 0.179	0.781 ± 0.099	0.692 ± 0.151	0.757 ± 0.314	0.476 ± 0.331	0.838 ± 0.111	0.778 ± 0.134	0.778 ± 0.042	–	–	–
Website Phishing	0.990 ± 0.042	0.913 ± 0.024	0.998 ± 0.007	0.936 ± 0.032	**1.000 ± 0.000**	0.997 ± 0.007	0.969 ± 0.064	0.997 ± 0.007	0.992 ± 0.013	0.882 ± 0.071	–	–	–

Table 19: Recall (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.921 ± 0.089	0.886 ± 0.037	0.031 ± 0.069	0.918 ± 0.042	**0.979**	0.879 ± 0.055	0.907 ± 0.094	0.669 ± 0.154	0.923 ± 0.049	0.975 ± 0.091	0.887	0.954	–
Breast Cancer	0.103 ± 0.102	0.651 ± 0.274	0.286 ± 0.066	0.895 ± 0.068	**0.960**	0.738 ± 0.152	0.588 ± 0.549	0.330 ± 0.109	0.913 ± 0.023	0.525 ± 0.210	–	–	–
Liver Disorders	0.918 ± 0.026	0.891 ± 0.199	0.563 ± 0.307	0.972 ± 0.005	0.971	0.762 ± 0.185	0.960 ± 0.080	0.199 ± 0.293	0.969 ± 0.045	**1.000 ± 0.000**	–	–	–
Parkinsons	0.372 ± 0.073	0.667 ± 0.016	0.434 ± 0.199	0.981 ± 0.000	**0.987**	0.684 ± 0.189	0.838 ± 0.151	0.363 ± 0.161	0.944 ± 0.115	0.447 ± 0.160	–	–	–
Obesity	0.923 ± 0.026	0.915 ± 0.020	0.351 ± 0.032	0.854 ± 0.197	0.933	0.907 ± 0.011	0.810 ± 0.145	0.881 ± 0.026	0.907 ± 0.059	**0.949 ± 0.054**	–	–	–
Diabetes	0.945 ± 0.063	0.963 ± 0.025	0.704 ± 0.143	0.973 ± 0.022	–	0.940 ± 0.070	0.910 ± 0.152	0.946 ± 0.056	0.964 ± 0.023	**0.988 ± 0.019**	0.856	–	–
German Credit	0.473 ± 1.310	0.907 ± 0.079	0.015 ± 0.002	0.932 ± 0.022	–	0.924 ± 0.035	0.917 ± 0.097	0.875 ± 0.005	0.940 ± 0.078	**0.988 ± 0.025**	–	–	–
Bank Marketing	0.790 ± 0.146	0.936 ± 0.016	0.298 ± 0.184	0.814 ± 0.161	–	0.840 ± 0.125	0.871 ± 0.072	0.775 ± 0.157	0.892 ± 0.103	**0.949 ± 0.184**	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Census Income	0.712 ± 0.170	0.911 ± 0.096	0.304 ± 0.172	0.883 ± 0.062	–	0.787 ± 0.036	0.631 ± 1.049	0.458 ± 0.739	0.840 ± 0.097	$\mathbf{0.917 \pm 0.265}$	–	–	–
Melanoma	0.947 ± 0.038	0.935 ± 0.159	0.423 ± 0.254	0.974 ± 0.023	0.990 ± 0.023	0.872 ± 0.215	0.929 ± 0.229	0.522 ± 0.273	0.986 ± 0.009	$\mathbf{1.000 \pm 0.000}$	–	–	–
PBC	0.782 ± 0.546	0.989 ± 0.030	0.553 ± 0.038	0.980 ± 0.010	0.993 ± 0.012	0.968 ± 0.043	0.968 ± 0.056	0.647 ± 0.709	0.972 ± 0.026	$\mathbf{1.000 \pm 0.000}$	–	–	–
PBC3	0.878 ± 0.073	0.910 ± 0.080	0.068 ± 0.039	0.946 ± 0.039	0.957 ± 0.058	0.842 ± 0.098	0.858 ± 0.298	0.455 ± 0.099	0.914 ± 0.041	$\mathbf{0.986 \pm 0.032}$	0.749	–	–
Retinopathy	0.668 ± 1.305	0.976 ± 0.044	0.244 ± 0.095	0.988 ± 0.034	$\mathbf{0.998 \pm 0.005}$	0.929 ± 0.022	0.965 ± 0.065	0.982 ± 0.042	0.976 ± 0.055	0.997 ± 0.015	–	–	–
WHAS500	0.684 ± 0.124	0.856 ± 0.263	0.047 ± 0.050	0.947 ± 0.014	0.957 ± 0.014	0.885 ± 0.110	0.906 ± 0.135	0.593 ± 0.238	0.925 ± 0.045	$\mathbf{1.000 \pm 0.000}$	–	–	–
GBSG2	0.888 ± 0.133	0.923 ± 0.098	0.190 ± 0.072	0.973 ± 0.014	0.957 ± 0.014	0.853 ± 0.142	0.904 ± 0.327	0.763 ± 0.053	0.964 ± 0.022	$\mathbf{0.985 \pm 0.032}$	0.880	–	–
Colon	0.948 ± 0.063	0.966 ± 0.040	0.259 ± 0.139	0.979 ± 0.013	0.983 ± 0.010	0.911 ± 0.139	0.788 ± 0.419	0.743 ± 0.028	0.978 ± 0.019	$\mathbf{0.995 \pm 0.017}$	–	–	–
GRACE	0.945 ± 0.041	0.965 ± 0.047	0.593 ± 0.454	0.982 ± 0.008	0.987 ± 0.016	0.811 ± 0.330	0.893 ± 0.419	0.299 ± 0.225	0.982 ± 0.006	$\mathbf{0.998 \pm 0.002}$	–	–	–
TRACE	0.950 ± 0.064	0.989 ± 0.003	0.476 ± 0.564	0.993 ± 0.008	0.990 ± 0.001	0.838 ± 0.191	0.977 ± 0.041	0.316 ± 0.097	0.993 ± 0.005	$\mathbf{1.000 \pm 0.001}$	–	–	–
Framingham	0.905 ± 0.016	0.974 ± 0.042	0.525 ± 0.025	0.970 ± 0.003	0.972 ± 0.008	0.904 ± 0.095	0.869 ± 0.253	0.663 ± 0.764	0.951 ± 0.026	$\mathbf{0.998 \pm 0.006}$	–	–	–
Blood Transfusion	0.613 ± 0.912	0.928 ± 0.120	0.583 ± 0.288	$\mathbf{0.995 \pm 0.009}$	0.992 ± 0.015	0.825 ± 0.166	0.928 ± 0.171	0.244 ± 0.098	0.980 ± 0.045	0.978 ± 0.055	–	–	–
Credit-G	0.188 ± 0.098	0.902 ± 0.044	0.036 ± 0.102	0.939 ± 0.021	0.914 ± 0.035	0.870 ± 0.093	0.806 ± 0.609	0.876 ± 0.072	0.914 ± 0.028	$\mathbf{0.976 \pm 0.017}$	–	–	–
QSAR Biodeg	0.278 ± 0.104	0.776 ± 0.066	0.141 ± 0.149	0.689 ± 0.111	$\mathbf{0.805 \pm 0.729}$	0.801 ± 0.106	0.119 ± 0.281	0.640 ± 0.014	0.772 ± 0.120	0.472 ± 1.128	–	–	–
Maternal Risk	0.912 ± 0.041	0.825 ± 0.298	0.154 ± 0.029	0.901 ± 0.196	0.962 ± 0.062	0.809 ± 0.260	0.807 ± 0.332	0.700 ± 0.072	0.953 ± 0.072	$\mathbf{0.972 \pm 0.095}$	–	–	–
Website Phishing	0.834 ± 0.687	0.836 ± 0.082	0.195 ± 0.103	0.826 ± 0.017	0.999 ± 0.002	$\mathbf{1.000 \pm 0.000}$	0.957 ± 0.168	$\mathbf{1.000 \pm 0.000}$	0.999 ± 0.005	$\mathbf{1.000 \pm 0.000}$	–	–	–

Table 20: Density (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.866 ± 0.521	0.489 ± 0.259	1.616 ± 0.212	0.904 ± 0.147	0.574	1.036 ± 0.204	0.250 ± 0.195	$\mathbf{1.623 \pm 0.132}$	0.942 ± 0.128	0.139 ± 0.040	1.092	1.152	–
Breast Cancer	0.311 ± 0.185	0.001 ± 0.003	0.078 ± 0.068	$\mathbf{0.590 \pm 0.106}$	0.507	0.135 ± 0.012	0.000 ± 0.001	0.556 ± 0.164	0.008 ± 0.016	0.011 ± 0.025	–	–	–
Liver Disorders	1.139 ± 0.220	0.268 ± 0.231	$\mathbf{1.375 \pm 0.251}$	0.946 ± 0.069	1.162	1.090 ± 0.271	0.370 ± 0.012	1.100 ± 0.161	0.874 ± 0.274	0.258 ± 0.097	–	–	–
Parkinsons	0.960 ± 0.632	0.001 ± 0.006	0.276 ± 0.178	0.575 ± 0.217	0.772	0.166 ± 0.143	0.003 ± 0.005	$\mathbf{0.992 \pm 0.094}$	0.054 ± 0.058	0.027 ± 0.032	–	–	–
Obesity	$\mathbf{0.904 \pm 0.034}$	0.225 ± 0.040	0.808 ± 0.126	0.286 ± 0.046	0.565	0.430 ± 0.399	0.153 ± 0.082	0.458 ± 0.060	0.252 ± 0.096	0.023 ± 0.027	–	–	–
Diabetes	1.021 ± 0.093	0.870 ± 0.156	$\mathbf{1.140 \pm 0.043}$	0.641 ± 0.120	–	0.941 ± 0.063	0.619 ± 0.783	0.836 ± 0.416	0.815 ± 0.133	0.213 ± 0.077	1.044	–	–
German Credit	0.742 ± 0.648	0.439 ± 0.132	$\mathbf{2.157 \pm 0.320}$	0.558 ± 0.014	–	0.784 ± 0.114	0.393 ± 0.095	0.911 ± 0.210	0.376 ± 0.051	0.039 ± 0.016	–	–	–
Bank Marketing	1.104 ± 0.343	0.449 ± 0.112	$\mathbf{1.565 \pm 0.044}$	0.347 ± 0.157	–	0.447 ± 0.163	0.147 ± 0.076	0.501 ± 0.476	0.319 ± 0.041	0.076 ± 0.150	–	–	–
Census Income	1.189 ± 0.481	0.328 ± 0.169	$\mathbf{1.667 \pm 0.184}$	0.156 ± 0.170	–	0.281 ± 0.381	0.055 ± 0.102	0.187 ± 0.351	0.132 ± 0.077	0.004 ± 0.011	–	–	–
Melanoma	1.107 ± 0.292	0.541 ± 0.088	1.111 ± 0.162	0.880 ± 0.170	0.981 ± 0.144	1.102 ± 0.087	0.512 ± 0.121	$\mathbf{1.292 \pm 0.174}$	0.923 ± 0.179	0.504 ± 0.131	–	–	–
PBC	0.777 ± 0.939	0.778 ± 0.173	$\mathbf{1.124 \pm 0.094}$	0.961 ± 0.102	1.056 ± 0.085	1.007 ± 0.066	0.779 ± 0.305	0.867 ± 1.303	1.016 ± 0.063	0.592 ± 0.136	–	–	–
PBC3	1.111 ± 0.614	0.344 ± 0.208	1.602 ± 0.237	0.746 ± 0.065	0.707 ± 0.183	0.881 ± 0.056	0.187 ± 0.100	$\mathbf{1.736 \pm 0.377}$	0.853 ± 0.235	0.066 ± 0.052	1.117	–	–
Retinopathy	0.706 ± 1.299	0.308 ± 0.036	$\mathbf{1.197 \pm 0.106}$	0.525 ± 0.122	1.112 ± 0.050	0.770 ± 0.443	0.341 ± 0.138	0.680 ± 0.047	0.508 ± 0.255	0.281 ± 0.122	–	–	–
WHAS500	1.457 ± 0.428	0.300 ± 0.216	1.686 ± 0.651	0.907 ± 0.141	1.010 ± 0.071	1.035 ± 0.207	0.152 ± 0.237	$\mathbf{1.693 \pm 0.512}$	0.889 ± 0.054	0.357 ± 0.281	–	–	–
GBSG2	0.870 ± 0.692	0.384 ± 0.225	$\mathbf{1.146 \pm 0.411}$	0.785 ± 0.085	0.995 ± 0.100	0.827 ± 0.514	0.304 ± 0.173	0.923 ± 0.108	0.728 ± 0.042	0.168 ± 0.090	0.869	–	–
Colon	1.096 ± 0.079	0.684 ± 0.087	$\mathbf{1.145 \pm 0.059}$	0.901 ± 0.013	1.060 ± 0.025	0.960 ± 0.116	0.372 ± 0.570	1.081 ± 0.013	0.893 ± 0.279	0.285 ± 0.047	–	–	–
GRACE	1.056 ± 0.122	0.693 ± 0.054	1.087 ± 0.099	0.951 ± 0.030	0.993 ± 0.052	1.012 ± 0.083	0.376 ± 0.806	$\mathbf{1.108 \pm 0.088}$	1.013 ± 0.043	0.360 ± 0.071	–	–	–
TRACE	$\mathbf{1.091 \pm 0.054}$	0.643 ± 0.118	0.982 ± 0.090	0.931 ± 0.047	1.024 ± 0.036	1.082 ± 0.115	0.525 ± 0.228	0.898 ± 0.838	1.023 ± 0.058	0.397 ± 0.022	–	–	–
Framingham	1.035 ± 0.123	0.391 ± 0.166	$\mathbf{1.136 \pm 0.102}$	0.871 ± 0.015	1.000 ± 0.007	0.823 ± 0.291	0.141 ± 0.272	0.498 ± 1.113	0.782 ± 0.154	0.098 ± 0.038	–	–	–
Blood Transfusion	0.537 ± 0.787	0.193 ± 0.057	0.687 ± 0.016	0.819 ± 0.157	$\mathbf{0.968 \pm 0.077}$	0.758 ± 0.137	0.338 ± 0.613	0.556 ± 0.052	0.574 ± 0.228	0.119 ± 0.043	–	–	–
Credit-G	0.436 ± 0.148	0.423 ± 0.092	$\mathbf{2.403 \pm 0.297}$	0.560 ± 0.040	1.180 ± 0.048	0.916 ± 0.278	0.390 ± 0.481	0.948 ± 0.050	0.457 ± 0.079	0.029 ± 0.006	–	–	–
QSAR Biodeg	0.043 ± 0.017	0.021 ± 0.003	$\mathbf{0.137 \pm 0.053}$	0.029 ± 0.017	0.035 ± 0.031	0.091 ± 0.030	0.001 ± 0.003	0.134 ± 0.031	0.021 ± 0.016	0.008 ± 0.003	–	–	–
Maternal Risk	$\mathbf{1.084 \pm 0.118}$	0.325 ± 0.269	0.984 ± 0.198	0.490 ± 0.104	0.403 ± 0.147	0.603 ± 0.727	0.238 ± 0.222	0.994 ± 0.262	0.591 ± 0.157	0.218 ± 0.034	–	–	–
Website Phishing	2.197 ± 2.674	0.839 ± 0.069	1.369 ± 0.228	0.884 ± 0.037	$\mathbf{2.753 \pm 0.046}$	2.272 ± 0.372	1.588 ± 0.531	2.224 ± 0.125	2.064 ± 0.492	0.979 ± 0.018	–	–	–

Table 21: Coverage (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.841 ± 0.320	0.681 ± 0.111	0.202 ± 0.097	0.968 ± 0.026	0.933	0.925 ± 0.027	0.400 ± 0.510	0.940 ± 0.026	0.961 ± 0.033	0.096 ± 0.000	0.941	**0.987**	–
Breast Cancer	0.217 ± 0.025	0.001 ± 0.003	0.144 ± 0.082	$\mathbf{0.789 \pm 0.055}$	0.778	0.304 ± 0.087	0.001 ± 0.003	0.305 ± 0.036	0.017 ± 0.025	0.008 ± 0.003	–	–	–
Liver Disorders	0.899 ± 0.039	0.349 ± 0.086	0.686 ± 0.218	0.961 ± 0.010	**0.989**	0.798 ± 0.090	0.566 ± 0.207	0.460 ± 0.289	0.929 ± 0.105	0.165 ± 0.034	–	–	–
Parkinsons	0.556 ± 0.096	0.004 ± 0.018	0.267 ± 0.079	0.812 ± 0.144	**0.872**	0.318 ± 0.199	0.015 ± 0.024	0.476 ± 0.133	0.171 ± 0.187	0.021 ± 0.009	–	–	–
Obesity	$\mathbf{0.882 \pm 0.070}$	0.304 ± 0.070	0.283 ± 0.014	0.295 ± 0.169	0.725	0.540 ± 0.233	0.221 ± 0.115	0.561 ± 0.031	0.354 ± 0.048	0.028 ± 0.004	–	–	–
Diabetes	$\mathbf{0.907 \pm 0.047}$	0.852 ± 0.024	0.763 ± 0.123	0.726 ± 0.154	–	0.902 ± 0.061	0.529 ± 0.218	0.861 ± 0.189	0.833 ± 0.081	0.164 ± 0.022	0.600	–	–
German Credit	0.783 ± 0.627	0.674 ± 0.127	0.310 ± 0.043	0.826 ± 0.006	–	0.905 ± 0.039	0.522 ± 0.128	$\mathbf{0.909 \pm 0.050}$	0.670 ± 0.065	0.076 ± 0.018	–	–	–
Bank Marketing	$\mathbf{0.898 \pm 0.058}$	0.693 ± 0.099	0.712 ± 0.146	0.581 ± 0.015	–	0.616 ± 0.163	0.237 ± 0.197	0.632 ± 0.369	0.502 ± 0.087	0.006 ± 0.008	–	–	–
Census Income	$\mathbf{0.841 \pm 0.081}$	0.527 ± 0.060	0.703 ± 0.154	0.288 ± 0.317	–	0.474 ± 0.425	0.100 ± 0.186	0.327 ± 0.512	0.306 ± 0.164	0.003 ± 0.000	–	–	–
Melanoma	0.911 ± 0.197	0.673 ± 0.106	0.453 ± 0.280	0.923 ± 0.032	$\mathbf{0.972 \pm 0.057}$	0.850 ± 0.229	0.516 ± 0.441	0.673 ± 0.166	0.931 ± 0.168	0.557 ± 0.155	–	–	–
PBC	0.477 ± 0.977	0.912 ± 0.098	0.562 ± 0.035	0.965 ± 0.025	$\mathbf{0.972 \pm 0.010}$	0.896 ± 0.095	0.778 ± 0.075	0.554 ± 0.781	0.940 ± 0.020	0.708 ± 0.120	–	–	–
PBC3	0.919 ± 0.105	0.596 ± 0.054	0.388 ± 0.063	0.897 ± 0.042	$\mathbf{0.944 \pm 0.058}$	0.872 ± 0.029	0.331 ± 0.294	0.888 ± 0.005	0.889 ± 0.079	0.082 ± 0.031	0.849	–	–
Retinopathy	0.702 ± 1.147	0.651 ± 0.123	0.308 ± 0.066	0.881 ± 0.034	$\mathbf{0.991 \pm 0.020}$	0.867 ± 0.250	0.591 ± 0.117	0.917 ± 0.020	0.773 ± 0.148	0.612 ± 0.228	–	–	–
WHAS500	0.915 ± 0.052	0.438 ± 0.301	0.271 ± 0.095	0.935 ± 0.036	$\mathbf{0.984 \pm 0.018}$	0.936 ± 0.016	0.281 ± 0.290	0.914 ± 0.068	0.894 ± 0.007	0.113 ± 0.044	–	–	–
GBSG2	0.814 ± 0.291	0.571 ± 0.188	0.220 ± 0.060	0.915 ± 0.025	$\mathbf{0.962 \pm 0.015}$	0.691 ± 0.396	0.411 ± 0.316	0.804 ± 0.046	0.881 ± 0.121	0.069 ± 0.016	0.743	–	–
Colon	0.958 ± 0.170	0.795 ± 0.129	0.321 ± 0.204	0.943 ± 0.036	$\mathbf{0.983 \pm 0.020}$	0.794 ± 0.301	0.399 ± 0.498	0.817 ± 0.036	0.935 ± 0.025	0.255 ± 0.073	–	–	–
GRACE	0.940 ± 0.040	0.662 ± 0.109	0.447 ± 0.174	0.954 ± 0.016	$\mathbf{0.970 \pm 0.008}$	0.739 ± 0.269	0.464 ± 0.996	0.481 ± 0.041	0.942 ± 0.035	0.447 ± 0.081	–	–	–
TRACE	0.933 ± 0.081	0.711 ± 0.105	0.314 ± 0.308	0.941 ± 0.033	$\mathbf{0.973 \pm 0.017}$	0.673 ± 0.156	0.519 ± 0.165	0.297 ± 0.427	0.890 ± 0.048	0.463 ± 0.053	–	–	–
Framingham	0.926 ± 0.034	0.611 ± 0.200	0.553 ± 0.144	0.942 ± 0.021	$\mathbf{0.970 \pm 0.007}$	0.844 ± 0.185	0.202 ± 0.072	0.405 ± 0.946	0.828 ± 0.036	0.138 ± 0.027	–	–	–
Blood Transfusion	0.376 ± 0.375	0.272 ± 0.172	0.347 ± 0.206	0.888 ± 0.059	$\mathbf{0.957 \pm 0.079}$	0.592 ± 0.222	0.282 ± 0.426	0.251 ± 0.157	0.785 ± 0.141	0.104 ± 0.022	–	–	–
Credit-G	0.623 ± 0.075	0.680 ± 0.081	0.335 ± 0.151	0.804 ± 0.064	$\mathbf{0.980 \pm 0.004}$	0.909 ± 0.044	0.422 ± 0.297	0.917 ± 0.032	0.735 ± 0.117	0.062 ± 0.010	–	–	–
QSAR Biodeg	0.085 ± 0.044	0.023 ± 0.006	0.086 ± 0.021	0.053 ± 0.027	0.001 ± 0.000	0.212 ± 0.065	0.003 ± 0.005	$\mathbf{0.248 \pm 0.042}$	0.048 ± 0.032	0.001 ± 0.000	–	–	–
Maternal Risk	$\mathbf{0.899 \pm 0.021}$	0.335 ± 0.079	0.245 ± 0.057	0.534 ± 0.112	0.675 ± 0.204	0.597 ± 0.484	0.270 ± 0.409	0.661 ± 0.068	0.715 ± 0.113	0.173 ± 0.046	–	–	–
Website Phishing	0.883 ± 0.440	0.751 ± 0.028	0.500 ± 0.007	0.788 ± 0.086	$\mathbf{1.000 \pm 0.000}$	0.988 ± 0.005	0.752 ± 0.176	0.982 ± 0.020	0.983 ± 0.013	0.838 ± 0.031	–	–	–

Detection / distribution

Table 22: C2ST AUROC (↓, 0.5 best) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.969 ± 0.039	0.956 ± 0.039	0.999 ± 0.003	0.902 ± 0.033	0.958	0.956 ± 0.014	0.976 ± 0.012	0.999 ± 0.003	0.939 ± 0.014	0.987 ± 0.003	0.873	**0.442**	–
Breast Cancer	1.000 ± 0.000	1.000 ± 0.000	0.993 ± 0.005	**0.817 ± 0.265**	0.828	0.989 ± 0.014	0.998 ± 0.001	1.000 ± 0.000	0.988 ± 0.008	0.998 ± 0.002	–	–	–
Liver Disorders	0.662 ± 0.068	0.960 ± 0.054	0.822 ± 0.062	0.593 ± 0.034	**0.461**	0.879 ± 0.055	0.914 ± 0.088	1.000 ± 0.000	0.648 ± 0.074	0.972 ± 0.002	–	–	–
Parkinsons	0.999 ± 0.005	0.995 ± 0.016	0.984 ± 0.012	0.732 ± 0.150	**0.728**	0.976 ± 0.021	0.993 ± 0.006	0.993 ± 0.016	0.951 ± 0.043	0.983 ± 0.010	–	–	–
Obesity	0.998 ± 0.002	0.993 ± 0.002	0.999 ± 0.001	0.991 ± 0.008	**0.839**	0.994 ± 0.003	0.997 ± 0.008	1.000 ± 0.000	0.993 ± 0.002	1.000 ± 0.000	–	–	–
Diabetes	1.000 ± 0.001	0.951 ± 0.022	0.927 ± 0.011	0.984 ± 0.002	–	0.999 ± 0.001	0.997 ± 0.002	1.000 ± 0.000	0.997 ± 0.002	0.997 ± 0.002	**0.841**	–	–
German Credit	1.000 ± 0.000	0.937 ± 0.035	0.999 ± 0.000	**0.877 ± 0.013**	–	0.892 ± 0.078	0.977 ± 0.013	1.000 ± 0.001	0.924 ± 0.011	0.996 ± 0.002	–	–	–
Bank Marketing	**0.888 ± 0.048**	0.910 ± 0.025	0.964 ± 0.023	0.985 ± 0.026	–	0.969 ± 0.062	0.998 ± 0.005	1.000 ± 0.000	0.968 ± 0.020	1.000 ± 0.000	–	–	–
Census Income	1.000 ± 0.000	**0.965 ± 0.020**	0.980 ± 0.014	0.990 ± 0.034	–	0.995 ± 0.004	0.998 ± 0.002	1.000 ± 0.001	0.993 ± 0.001	1.000 ± 0.000	–	–	–
Melanoma	0.853 ± 0.047	0.884 ± 0.098	0.924 ± 0.024	**0.691 ± 0.121**	0.726 ± 0.076	0.841 ± 0.123	0.922 ± 0.147	0.999 ± 0.002	0.717 ± 0.075	0.890 ± 0.077	–	–	–
PBC	0.729 ± 0.575	0.629 ± 0.084	0.839 ± 0.092	0.499 ± 0.148	**0.494 ± 0.017**	0.633 ± 0.068	0.790 ± 0.093	0.875 ± 0.458	0.536 ± 0.115	0.832 ± 0.045	–	–	–
PBC3	0.913 ± 0.005	0.944 ± 0.070	0.987 ± 0.002	**0.802 ± 0.076**	0.892 ± 0.009	0.912 ± 0.017	0.970 ± 0.064	1.000 ± 0.000	0.849 ± 0.014	0.986 ± 0.001	0.806	–	–
Retinopathy	0.744 ± 0.580	0.957 ± 0.039	0.939 ± 0.054	0.831 ± 0.063	**0.494 ± 0.061**	0.795 ± 0.202	0.948 ± 0.023	0.981 ± 0.014	0.881 ± 0.065	0.944 ± 0.012	–	–	–
WHAS500	1.000 ± 0.000	0.960 ± 0.076	0.991 ± 0.008	**0.568 ± 0.063**	0.713 ± 0.100	0.785 ± 0.049	0.971 ± 0.066	1.000 ± 0.000	0.675 ± 0.032	0.985 ± 0.004	–	–	–
GBSG2	0.857 ± 0.256	0.941 ± 0.051	0.987 ± 0.008	0.788 ± 0.025	**0.675 ± 0.027**	0.914 ± 0.151	0.959 ± 0.066	1.000 ± 0.000	0.767 ± 0.045	0.994 ± 0.002	0.845	–	–
Colon	**0.600 ± 0.062**	0.854 ± 0.053	0.943 ± 0.060	0.652 ± 0.037	0.776 ± 0.008	0.994 ± 0.003	0.997 ± 0.004	1.000 ± 0.000	0.995 ± 0.004	0.989 ± 0.005	–	–	–
GRACE	0.671 ± 0.038	0.885 ± 0.055	0.927 ± 0.042	0.697 ± 0.027	**0.618 ± 0.104**	0.867 ± 0.080	0.927 ± 0.167	1.000 ± 0.000	0.734 ± 0.040	0.942 ± 0.018	–	–	–
TRACE	0.999 ± 0.002	0.851 ± 0.083	0.926 ± 0.083	**0.614 ± 0.021**	0.664 ± 0.045	0.995 ± 0.003	0.992 ± 0.002	0.999 ± 0.003	0.993 ± 0.003	0.985 ± 0.004	–	–	–
Framingham	0.804 ± 0.036	0.914 ± 0.062	0.952 ± 0.015	0.771 ± 0.011	**0.650 ± 0.042**	0.910 ± 0.081	0.988 ± 0.003	0.996 ± 0.012	0.864 ± 0.031	0.988 ± 0.004	–	–	–
Blood Transfusion	0.992 ± 0.027	0.989 ± 0.017	0.996 ± 0.001	0.969 ± 0.030	**0.933 ± 0.063**	0.986 ± 0.010	0.989 ± 0.014	1.000 ± 0.000	0.979 ± 0.016	0.992 ± 0.005	–	–	–
Credit-G	1.000 ± 0.000	0.942 ± 0.016	0.999 ± 0.001	0.870 ± 0.051	**0.688 ± 0.053**	0.894 ± 0.044	0.977 ± 0.057	1.000 ± 0.001	0.920 ± 0.014	0.997 ± 0.001	–	–	–
QSAR Biodeg	1.000 ± 0.002	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	**0.999 ± 0.004**	0.999 ± 0.001	0.999 ± 0.003	1.000 ± 0.000	0.999 ± 0.001	1.000 ± 0.000	–	–	–
Maternal Risk	0.999 ± 0.002	0.998 ± 0.001	0.997 ± 0.002	0.992 ± 0.004	**0.977 ± 0.020**	0.993 ± 0.004	0.995 ± 0.008	0.999 ± 0.004	0.983 ± 0.002	0.995 ± 0.003	–	–	–
Website Phishing	0.771 ± 0.501	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	**0.363 ± 0.059**	0.635 ± 0.052	0.927 ± 0.034	0.664 ± 0.047	0.707 ± 0.081	0.904 ± 0.029	–	–	–

Table 23: MMD^2 (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.050 ± 0.074	0.099 ± 0.049	0.392 ± 0.088	**0.000 ± 0.000**	0.188	0.033 ± 0.010	0.242 ± 0.210	0.027 ± 0.007	0.007 ± 0.011	0.399 ± 0.025	0.011	0.000	–
Breast Cancer	0.104 ± 0.055	0.218 ± 0.041	0.110 ± 0.068	**0.009 ± 0.003**	0.227	0.058 ± 0.013	0.806 ± 0.460	0.101 ± 0.028	0.039 ± 0.057	0.573 ± 0.050	–	–	–
Liver Disorders	0.051 ± 0.056	0.276 ± 0.126	0.118 ± 0.148	0.001 ± 0.002	**0.000**	0.066 ± 0.048	0.132 ± 0.156	0.066 ± 0.030	0.021 ± 0.034	0.480 ± 0.041	–	–	–
Parkinsons	0.083 ± 0.117	0.202 ± 0.086	0.140 ± 0.050	**0.000 ± 0.000**	0.121	0.096 ± 0.059	0.145 ± 0.265	0.119 ± 0.032	0.022 ± 0.016	0.488 ± 0.007	–	–	–
Obesity	**0.005 ± 0.006**	0.037 ± 0.027	0.081 ± 0.022	0.031 ± 0.042	0.014	0.008 ± 0.020	0.109 ± 0.200	0.006 ± 0.002	0.019 ± 0.010	0.202 ± 0.030	–	–	–
Diabetes	0.024 ± 0.072	0.019 ± 0.007	0.017 ± 0.019	0.019 ± 0.030	–	**0.007 ± 0.007**	0.157 ± 0.159	0.025 ± 0.086	0.019 ± 0.019	0.381 ± 0.085	0.057	–	–
German Credit	0.046 ± 0.027	0.024 ± 0.030	0.299 ± 0.013	**0.001 ± 0.003**	–	0.008 ± 0.009	0.049 ± 0.039	0.007 ± 0.004	0.015 ± 0.010	0.238 ± 0.022	–	–	–
Bank Marketing	0.012 ± 0.012	**0.010 ± 0.006**	0.033 ± 0.014	0.012 ± 0.009	–	0.021 ± 0.015	0.090 ± 0.130	0.019 ± 0.017	0.027 ± 0.018	0.873 ± 0.104	–	–	–
Census Income	0.039 ± 0.047	**0.012 ± 0.010**	0.039 ± 0.040	0.123 ± 0.294	–	0.048 ± 0.057	0.224 ± 0.647	0.065 ± 0.111	0.044 ± 0.032	0.676 ± 0.035	–	–	–
Melanoma	0.063 ± 0.192	0.153 ± 0.073	0.199 ± 0.186	0.005 ± 0.014	**0.001 ± 0.002**	0.036 ± 0.064	0.223 ± 0.252	0.031 ± 0.043	0.008 ± 0.013	0.179 ± 0.051	–	–	–
PBC	0.038 ± 0.054	0.028 ± 0.044	0.139 ± 0.084	0.004 ± 0.013	**0.000 ± 0.000**	0.029 ± 0.032	0.106 ± 0.069	0.109 ± 0.386	0.024 ± 0.023	0.153 ± 0.025	–	–	–
PBC3	0.026 ± 0.021	0.069 ± 0.053	0.194 ± 0.066	**0.000 ± 0.001**	0.107 ± 0.070	0.040 ± 0.015	0.225 ± 0.202	0.049 ± 0.023	0.016 ± 0.009	0.340 ± 0.021	0.027	–	–
Retinopathy	0.042 ± 0.129	0.029 ± 0.062	0.171 ± 0.060	0.006 ± 0.005	**0.000 ± 0.000**	0.018 ± 0.035	0.052 ± 0.049	0.007 ± 0.006	0.028 ± 0.019	0.045 ± 0.019	–	–	–
WHAS500	0.027 ± 0.036	0.115 ± 0.092	0.273 ± 0.110	**0.000 ± 0.001**	0.022 ± 0.026	0.017 ± 0.021	0.259 ± 0.275	0.034 ± 0.043	0.007 ± 0.003	0.394 ± 0.016	–	–	–
GBSG2	0.038 ± 0.070	0.107 ± 0.116	0.325 ± 0.082	0.007 ± 0.010	**0.002 ± 0.002**	0.087 ± 0.190	0.200 ± 0.235	0.020 ± 0.008	0.006 ± 0.008	0.661 ± 0.073	0.039	–	–
Colon	0.007 ± 0.012	0.035 ± 0.044	0.190 ± 0.145	0.001 ± 0.002	**0.000 ± 0.000**	0.052 ± 0.097	0.187 ± 0.228	0.009 ± 0.010	0.007 ± 0.005	0.386 ± 0.055	–	–	–
GRACE	0.013 ± 0.029	0.134 ± 0.121	0.181 ± 0.110	**0.001 ± 0.001**	0.001 ± 0.002	0.059 ± 0.072	0.422 ± 1.087	0.011 ± 0.001	0.011 ± 0.020	0.271 ± 0.073	–	–	–
TRACE	0.008 ± 0.028	0.103 ± 0.068	0.241 ± 0.248	0.001 ± 0.001	**0.000 ± 0.000**	0.017 ± 0.020	0.213 ± 0.094	0.140 ± 0.430	0.013 ± 0.010	0.232 ± 0.022	–	–	–
Framingham	0.007 ± 0.002	0.065 ± 0.048	0.073 ± 0.078	0.001 ± 0.001	**0.000 ± 0.000**	0.027 ± 0.036	0.380 ± 0.213	0.188 ± 0.399	0.019 ± 0.018	0.365 ± 0.077	–	–	–
Blood Transfusion	0.108 ± 0.355	0.225 ± 0.078	0.275 ± 0.200	0.009 ± 0.024	**0.000 ± 0.001**	0.089 ± 0.049	0.292 ± 0.265	0.026 ± 0.017	0.014 ± 0.017	0.632 ± 0.039	–	–	–
Credit-G	0.085 ± 0.016	0.026 ± 0.048	0.296 ± 0.071	**0.001 ± 0.002**	0.001 ± 0.002	0.010 ± 0.017	0.079 ± 0.127	0.006 ± 0.003	0.014 ± 0.007	0.244 ± 0.031	–	–	–
QSAR Biodeg	0.067 ± 0.073	0.110 ± 0.011	0.189 ± 0.041	0.149 ± 0.153	0.572 ± 0.052	0.034 ± 0.027	0.568 ± 0.387	**0.025 ± 0.016**	0.036 ± 0.056	0.680 ± 0.030	–	–	–
Maternal Risk	**0.007 ± 0.010**	0.141 ± 0.085	0.230 ± 0.151	0.015 ± 0.011	0.151 ± 0.183	0.066 ± 0.165	0.316 ± 0.525	0.019 ± 0.023	0.017 ± 0.019	0.360 ± 0.064	–	–	–
Website Phishing	0.043 ± 0.155	0.093 ± 0.001	0.138 ± 0.018	0.091 ± 0.002	**0.000 ± 0.000**	0.005 ± 0.009	0.083 ± 0.069	0.003 ± 0.005	0.009 ± 0.008	0.034 ± 0.006	–	–	–

Table 24: Fréchet-PCA (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	2.197 ± 1.768	4.871 ± 3.551	10.777 ± 1.890	**0.309 ± 0.172**	36.258	1.474 ± 0.435	11.078 ± 7.624	1.620 ± 0.978	0.651 ± 0.427	36.526 ± 6.670	1.810	0.357	–
Breast Cancer	24.230 ± 35.796	29.409 ± 5.766	13.849 ± 4.611	**0.790 ± 0.154**	262.006	5.478 ± 2.510	107.847 ± 59.046	12.971 ± 1.348	11.090 ± 12.826	216.031 ± 50.704	–	–	–
Liver Disorders	0.799 ± 0.631	5.592 ± 4.179	2.911 ± 0.827	**0.102 ± 0.040**	0.163	1.003 ± 1.006	3.041 ± 2.091	1.369 ± 0.855	0.689 ± 0.564	26.046 ± 2.381	–	–	–
Parkinsons	6.899 ± 3.221	20.827 ± 6.418	12.444 ± 2.154	**0.514 ± 0.286**	77.241	7.191 ± 3.318	19.322 ± 15.333	12.159 ± 1.849	5.990 ± 2.586	83.740 ± 5.762	–	–	–
Obesity	**0.287 ± 0.337**	1.640 ± 0.992	2.601 ± 0.085	1.213 ± 1.935	2.692	0.349 ± 0.768	4.702 ± 5.667	0.368 ± 0.063	0.948 ± 0.500	10.496 ± 2.717	–	–	–
Diabetes	1.776 ± 5.212	1.105 ± 0.353	0.953 ± 0.723	0.542 ± 0.953	–	**0.263 ± 0.077**	3.167 ± 1.559	0.796 ± 1.901	0.996 ± 1.917	27.853 ± 17.394	1.565	–	–
German Credit	1.858 ± 5.398	1.687 ± 1.583	8.982 ± 0.810	**0.254 ± 0.129**	–	0.435 ± 0.334	2.723 ± 1.713	0.581 ± 0.146	1.105 ± 0.711	16.535 ± 1.822	–	–	–
Bank Marketing	0.856 ± 0.799	**0.585 ± 0.580**	2.557 ± 1.284	3.818 ± 12.599	–	1.044 ± 0.846	12.075 ± 30.720	0.902 ± 0.377	8.494 ± 29.701	1835.097 ± 5924.108	–	–	–
Census Income	1.344 ± 1.265	**0.799 ± 0.778**	3.091 ± 1.218	21.677 ± 39.564	–	1.784 ± 1.003	7.884 ± 15.023	2.709 ± 1.327	1.540 ± 1.072	113.086 ± 33.070	–	–	–
Melanoma	0.975 ± 2.300	3.143 ± 2.444	3.381 ± 1.659	**0.352 ± 0.782**	0.543 ± 0.510	0.801 ± 1.452	4.625 ± 4.366	0.859 ± 0.984	0.386 ± 0.476	4.857 ± 2.184	–	–	–
PBC	1.520 ± 2.591	0.685 ± 0.733	3.424 ± 0.507	0.234 ± 0.485	**0.062 ± 0.049**	0.475 ± 0.287	1.865 ± 0.658	2.770 ± 9.431	0.439 ± 0.332	4.204 ± 0.987	–	–	–
PBC3	1.247 ± 0.927	3.377 ± 2.267	6.796 ± 1.308	**0.410 ± 0.261**	21.966 ± 14.353	1.710 ± 1.121	11.014 ± 5.263	2.785 ± 0.549	1.009 ± 0.265	28.665 ± 2.528	1.802	–	–
Retinopathy	0.982 ± 2.703	1.901 ± 1.251	3.689 ± 0.839	1.010 ± 0.404	**0.071 ± 0.089**	0.824 ± 0.867	2.196 ± 1.095	0.966 ± 0.197	1.526 ± 0.668	2.611 ± 0.508	–	–	–
WHAS500	1.544 ± 1.357	7.186 ± 5.019	10.013 ± 2.041	**0.309 ± 0.074**	10.632 ± 13.606	0.955 ± 0.983	16.775 ± 10.154	1.925 ± 1.097	0.928 ± 1.290	67.247 ± 23.059	–	–	–
GBSG2	11.501 ± 46.501	2.908 ± 2.172	5.838 ± 0.697	0.546 ± 0.458	2.622 ± 3.262	2.080 ± 3.255	8.209 ± 14.802	1.058 ± 0.462	**0.470 ± 0.379**	62.397 ± 16.244	0.931	–	–
Colon	0.417 ± 1.688	1.440 ± 1.317	6.550 ± 1.731	0.199 ± 0.096	**0.136 ± 0.118**	1.398 ± 2.134	5.128 ± 3.767	0.730 ± 0.497	0.493 ± 0.805	26.894 ± 8.398	–	–	–
GRACE	0.219 ± 0.264	2.794 ± 2.613	2.638 ± 1.011	**0.071 ± 0.021**	0.183 ± 0.247	0.998 ± 0.921	9.721 ± 23.582	0.345 ± 0.082	0.194 ± 0.296	11.076 ± 5.627	–	–	–
TRACE	0.308 ± 1.020	2.583 ± 1.281	5.195 ± 2.293	0.072 ± 0.042	**0.012 ± 0.007**	0.659 ± 0.398	3.817 ± 2.281	3.200 ± 8.118	0.410 ± 0.376	8.624 ± 0.478	–	–	–
Framingham	0.209 ± 0.105	2.388 ± 1.180	1.777 ± 1.239	**0.021 ± 0.020**	0.125 ± 0.254	0.623 ± 0.708	10.909 ± 10.687	4.702 ± 9.962	0.590 ± 0.403	29.902 ± 8.845	–	–	–
Blood Transfusion	1.560 ± 3.196	3.815 ± 0.776	3.424 ± 1.300	0.236 ± 0.581	**0.178 ± 0.298**	1.406 ± 0.086	3.423 ± 2.467	1.012 ± 0.429	0.921 ± 0.542	38.142 ± 4.669	–	–	–
Credit-G	2.865 ± 0.397	1.781 ± 2.503	8.753 ± 2.528	**0.248 ± 0.095**	0.357 ± 0.169	0.532 ± 0.671	3.872 ± 4.041	0.574 ± 0.243	0.919 ± 0.376	17.004 ± 2.819	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
QSAR Biodeg	14.141 ± 1.739	17.631 ± 3.741	20.915 ± 2.773	22.391 ± 27.121	1013.505 ± 203.490	**4.580 ± 1.868**	59.061 ± 54.876	9.061 ± 5.276	10.922 ± 9.358	754.443 ± 102.561	–	–	–
Maternal Risk	**0.275 ± 0.121**	4.425 ± 2.743	4.762 ± 0.952	0.675 ± 0.925	16.795 ± 19.484	1.322 ± 3.054	5.393 ± 6.872	0.583 ± 0.543	0.632 ± 0.218	18.273 ± 4.293	–	–	–
Website Phishing	0.562 ± 1.887	1.231 ± 0.144	2.260 ± 0.182	1.212 ± 0.084	**0.024 ± 0.019**	0.119 ± 0.090	1.272 ± 0.917	0.099 ± 0.087	0.198 ± 0.118	0.612 ± 0.098	–	–	–

Table 25: Wasserstein (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.319 ± 0.238	0.391 ± 0.066	0.597 ± 0.056	0.111 ± 0.027	0.958	0.247 ± 0.038	0.685 ± 0.228	0.308 ± 0.058	0.142 ± 0.063	1.085 ± 0.106	0.176	**0.094**	–
Breast Cancer	0.309 ± 0.054	0.692 ± 0.122	0.393 ± 0.103	**0.150 ± 0.026**	1.851	0.309 ± 0.043	1.567 ± 0.561	0.510 ± 0.065	0.167 ± 0.172	2.098 ± 0.235	–	–	–
Liver Disorders	0.378 ± 0.237	0.863 ± 0.148	0.423 ± 0.139	**0.109 ± 0.024**	0.113	0.356 ± 0.066	0.486 ± 0.236	0.425 ± 0.089	0.236 ± 0.149	1.872 ± 0.095	–	–	–
Parkinsons	0.264 ± 0.139	0.622 ± 0.122	0.406 ± 0.061	**0.120 ± 0.014**	1.227	0.384 ± 0.126	0.471 ± 0.482	0.540 ± 0.058	0.164 ± 0.045	1.477 ± 0.048	–	–	–
Obesity	**0.119 ± 0.042**	0.308 ± 0.151	0.358 ± 0.054	0.296 ± 0.123	0.300	0.172 ± 0.133	0.539 ± 0.618	0.175 ± 0.016	0.187 ± 0.091	0.727 ± 0.069	–	–	–
Diabetes	0.235 ± 0.361	0.245 ± 0.016	0.164 ± 0.063	0.184 ± 0.096	–	**0.144 ± 0.036**	0.471 ± 0.165	0.185 ± 0.196	0.207 ± 0.098	1.483 ± 0.399	0.230	–	–
German Credit	0.225 ± 0.341	0.252 ± 0.099	0.663 ± 0.091	**0.077 ± 0.028**	–	0.157 ± 0.070	0.406 ± 0.157	0.185 ± 0.024	0.190 ± 0.115	1.113 ± 0.097	–	–	–
Bank Marketing	0.162 ± 0.050	**0.133 ± 0.066**	0.245 ± 0.048	0.272 ± 0.184	–	0.184 ± 0.067	0.648 ± 0.882	0.201 ± 0.019	0.221 ± 0.179	7.511 ± 8.825	–	–	–
Census Income	0.279 ± 0.245	**0.141 ± 0.084**	0.264 ± 0.113	0.762 ± 1.192	–	0.248 ± 0.047	0.623 ± 0.925	0.202 ± 0.050	0.194 ± 0.096	3.209 ± 0.490	–	–	–
Melanoma	0.327 ± 0.570	0.580 ± 0.235	0.547 ± 0.213	**0.132 ± 0.109**	0.176 ± 0.078	0.273 ± 0.278	0.726 ± 0.538	0.333 ± 0.108	0.183 ± 0.068	0.686 ± 0.139	–	–	–
PBC	0.262 ± 0.135	0.195 ± 0.066	0.406 ± 0.095	0.090 ± 0.126	**0.055 ± 0.026**	0.226 ± 0.064	0.417 ± 0.154	0.354 ± 0.853	0.215 ± 0.095	0.510 ± 0.041	–	–	–
PBC3	0.227 ± 0.144	0.380 ± 0.131	0.422 ± 0.057	**0.098 ± 0.043**	0.786 ± 0.324	0.290 ± 0.122	0.718 ± 0.237	0.385 ± 0.055	0.168 ± 0.022	1.194 ± 0.035	0.214	–	–
Retinopathy	0.164 ± 0.153	0.191 ± 0.259	0.622 ± 0.134	0.109 ± 0.070	**0.052 ± 0.017**	0.204 ± 0.179	0.305 ± 0.222	0.193 ± 0.075	0.215 ± 0.153	0.323 ± 0.024	–	–	–
WHAS500	0.223 ± 0.155	0.435 ± 0.195	0.491 ± 0.065	**0.070 ± 0.016**	0.312 ± 0.171	0.176 ± 0.083	0.615 ± 0.359	0.298 ± 0.126	0.127 ± 0.031	1.128 ± 0.056	–	–	–
GBSG2	0.270 ± 0.367	0.524 ± 0.172	0.547 ± 0.110	0.172 ± 0.069	0.231 ± 0.153	0.405 ± 0.369	0.801 ± 0.599	0.321 ± 0.052	**0.124 ± 0.032**	2.844 ± 0.265	0.242	–	–
Colon	0.137 ± 0.011	0.219 ± 0.080	0.448 ± 0.125	0.072 ± 0.046	**0.050 ± 0.012**	0.268 ± 0.234	0.501 ± 0.337	0.166 ± 0.051	0.117 ± 0.081	1.267 ± 0.162	–	–	–
GRACE	0.159 ± 0.104	0.477 ± 0.248	0.521 ± 0.116	**0.075 ± 0.026**	0.100 ± 0.043	0.336 ± 0.199	0.967 ± 1.322	0.247 ± 0.036	0.153 ± 0.104	0.941 ± 0.126	–	–	–
TRACE	0.099 ± 0.143	0.414 ± 0.163	0.487 ± 0.212	0.048 ± 0.034	**0.032 ± 0.007**	0.209 ± 0.034	0.547 ± 0.186	0.432 ± 0.568	0.175 ± 0.088	0.885 ± 0.017	–	–	–
Framingham	0.135 ± 0.036	0.398 ± 0.211	0.280 ± 0.059	0.056 ± 0.024	**0.037 ± 0.023**	0.232 ± 0.127	0.985 ± 0.375	0.581 ± 0.872	0.214 ± 0.130	1.341 ± 0.154	–	–	–
Blood Transfusion	0.397 ± 0.744	0.636 ± 0.264	0.577 ± 0.137	0.189 ± 0.202	**0.117 ± 0.033**	0.459 ± 0.094	0.658 ± 0.258	0.323 ± 0.090	0.167 ± 0.116	2.406 ± 0.154	–	–	–
Credit-G	0.178 ± 0.076	0.241 ± 0.212	0.659 ± 0.123	**0.064 ± 0.025**	0.102 ± 0.038	0.177 ± 0.186	0.464 ± 0.445	0.194 ± 0.033	0.173 ± 0.042	1.133 ± 0.086	–	–	–
QSAR Biodeg	0.281 ± 0.076	0.398 ± 0.018	0.343 ± 0.027	0.364 ± 0.090	3.165 ± 0.165	0.192 ± 0.058	0.787 ± 0.514	0.244 ± 0.025	**0.184 ± 0.084**	3.328 ± 0.200	–	–	–
Maternal Risk	**0.146 ± 0.084**	0.615 ± 0.273	0.559 ± 0.072	0.255 ± 0.075	0.933 ± 0.560	0.323 ± 0.394	0.694 ± 0.565	0.304 ± 0.062	0.174 ± 0.060	1.182 ± 0.096	–	–	–
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 26: Association error (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.057 ± 0.005	0.059 ± 0.005	0.070 ± 0.006	0.055 ± 0.007	0.150	0.069 ± 0.012	0.094 ± 0.022	0.060 ± 0.012	0.058 ± 0.008	0.063 ± 0.009	0.064	**0.046**	–
Breast Cancer	0.278 ± 0.019	0.347 ± 0.010	0.195 ± 0.007	**0.051 ± 0.017**	0.188	0.107 ± 0.028	0.290 ± 0.050	0.240 ± 0.022	0.298 ± 0.045	0.357 ± 0.012	–	–	–
Liver Disorders	0.077 ± 0.095	0.222 ± 0.028	0.156 ± 0.099	**0.052 ± 0.005**	0.080	0.118 ± 0.122	0.216 ± 0.077	0.207 ± 0.025	0.171 ± 0.019	0.232 ± 0.048	–	–	–
Parkinsons	0.242 ± 0.046	0.444 ± 0.026	0.335 ± 0.008	**0.056 ± 0.018**	0.224	0.175 ± 0.029	0.366 ± 0.025	0.369 ± 0.044	0.378 ± 0.054	0.450 ± 0.016	–	–	–
Obesity	0.047 ± 0.002	0.103 ± 0.004	0.086 ± 0.008	0.069 ± 0.006	0.115	**0.039 ± 0.058**	0.091 ± 0.017	0.074 ± 0.008	0.087 ± 0.013	0.105 ± 0.002	–	–	–
Diabetes	0.055 ± 0.056	0.090 ± 0.048	0.056 ± 0.028	0.071 ± 0.014	–	**0.027 ± 0.003**	0.079 ± 0.027	0.075 ± 0.018	0.065 ± 0.010	0.098 ± 0.006	0.054	–	–
German Credit	0.108 ± 0.509	0.085 ± 0.004	0.111 ± 0.014	0.072 ± 0.006	–	**0.048 ± 0.010**	0.082 ± 0.003	0.063 ± 0.013	0.080 ± 0.004	0.086 ± 0.003	–	–	–
Bank Marketing	0.038 ± 0.004	0.075 ± 0.021	0.079 ± 0.023	0.067 ± 0.004	–	**0.034 ± 0.014**	0.084 ± 0.009	0.043 ± 0.023	0.072 ± 0.005	0.089 ± 0.003	–	–	–
Census Income	0.059 ± 0.024	0.081 ± 0.007	0.075 ± 0.013	0.108 ± 0.017	–	**0.044 ± 0.011**	0.116 ± 0.015	0.070 ± 0.036	0.104 ± 0.007	0.127 ± 0.002	–	–	–
Melanoma	**0.074 ± 0.023**	0.127 ± 0.027	0.108 ± 0.033	0.097 ± 0.053	0.096 ± 0.023	0.086 ± 0.068	0.119 ± 0.076	0.113 ± 0.023	0.109 ± 0.086	0.130 ± 0.008	–	–	–
PBC	0.109 ± 0.112	0.089 ± 0.021	0.137 ± 0.042	0.078 ± 0.033	**0.053 ± 0.016**	0.060 ± 0.045	0.087 ± 0.016	0.082 ± 0.046	0.073 ± 0.026	0.090 ± 0.037	–	–	–
PBC3	0.081 ± 0.019	0.101 ± 0.018	0.104 ± 0.019	0.071 ± 0.017	0.101 ± 0.036	**0.068 ± 0.034**	0.099 ± 0.013	0.086 ± 0.009	0.096 ± 0.002	0.097 ± 0.010	0.106	–	–
Retinopathy	0.082 ± 0.086	0.118 ± 0.024	0.208 ± 0.145	0.092 ± 0.031	**0.041 ± 0.036**	0.088 ± 0.029	0.120 ± 0.010	0.093 ± 0.022	0.097 ± 0.038	0.120 ± 0.007	–	–	–
WHAS500	0.069 ± 0.020	0.074 ± 0.005	0.089 ± 0.007	**0.054 ± 0.007**	0.088 ± 0.031	0.055 ± 0.015	0.084 ± 0.029	0.064 ± 0.005	0.066 ± 0.009	0.071 ± 0.004	–	–	–
GBSG2	**0.062 ± 0.042**	0.124 ± 0.017	0.104 ± 0.018	0.077 ± 0.019	0.078 ± 0.063	0.098 ± 0.088	0.126 ± 0.008	0.090 ± 0.043	0.092 ± 0.036	0.121 ± 0.030	0.091	–	–
Colon	0.038 ± 0.017	0.053 ± 0.002	0.081 ± 0.015	0.044 ± 0.011	**0.036 ± 0.005**	0.082 ± 0.055	0.061 ± 0.013	0.043 ± 0.013	0.049 ± 0.015	0.057 ± 0.003	–	–	–
GRACE	0.039 ± 0.020	0.061 ± 0.017	0.108 ± 0.063	**0.037 ± 0.004**	0.048 ± 0.056	0.044 ± 0.018	0.088 ± 0.060	0.060 ± 0.023	0.040 ± 0.039	0.055 ± 0.010	–	–	–
TRACE	0.034 ± 0.019	0.102 ± 0.008	0.077 ± 0.022	0.064 ± 0.007	**0.017 ± 0.006**	0.062 ± 0.053	0.095 ± 0.030	0.104 ± 0.045	0.066 ± 0.040	0.109 ± 0.007	–	–	–
Framingham	0.023 ± 0.014	0.136 ± 0.022	0.259 ± 0.052	**0.014 ± 0.006**	0.029 ± 0.021	0.032 ± 0.005	0.144 ± 0.045	0.107 ± 0.003	0.113 ± 0.020	0.152 ± 0.008	–	–	–
Blood Transfusion	0.233 ± 0.628	0.412 ± 0.027	0.250 ± 0.080	0.069 ± 0.038	**0.031 ± 0.051**	0.120 ± 0.110	0.369 ± 0.086	0.315 ± 0.176	0.266 ± 0.145	0.402 ± 0.057	–	–	–
Credit-G	0.064 ± 0.003	0.086 ± 0.006	0.110 ± 0.010	0.073 ± 0.020	0.064 ± 0.025	**0.047 ± 0.008**	0.081 ± 0.012	0.061 ± 0.007	0.077 ± 0.006	0.086 ± 0.003	–	–	–
QSAR Biodeg	0.154 ± 0.022	0.147 ± 0.004	0.128 ± 0.018	**0.079 ± 0.009**	0.131 ± 0.012	0.083 ± 0.009	0.147 ± 0.019	0.117 ± 0.001	0.142 ± 0.008	0.147 ± 0.003	–	–	–
Maternal Risk	0.079 ± 0.068	0.194 ± 0.022	0.142 ± 0.033	**0.071 ± 0.017**	0.158 ± 0.077	0.081 ± 0.037	0.166 ± 0.044	0.114 ± 0.044	0.160 ± 0.034	0.205 ± 0.010	–	–	–
Website Phishing	0.081 ± 0.082	0.100 ± 0.009	0.187 ± 0.072	0.091 ± 0.012	**0.030 ± 0.006**	0.058 ± 0.013	0.105 ± 0.005	0.076 ± 0.012	0.067 ± 0.014	0.106 ± 0.008	–	–	–

SDMetrics quality report

Table 27: Overall quality (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.374 ± 0.062	0.369 ± 0.021	0.307 ± 0.016	0.423 ± 0.010	0.341	0.387 ± 0.010	0.297 ± 0.060	0.348 ± 0.017	0.415 ± 0.021	0.317 ± 0.015	0.407	**0.438**	–
Breast Cancer	0.707 ± 0.025	0.600 ± 0.005	0.719 ± 0.033	**0.849 ± 0.004**	0.704	0.783 ± 0.017	0.404 ± 0.149	0.614 ± 0.031	0.747 ± 0.003	0.522 ± 0.009	–	–	–
Liver Disorders	0.805 ± 0.080	0.597 ± 0.056	0.705 ± 0.040	**0.870 ± 0.003**	0.866	0.787 ± 0.024	0.635 ± 0.104	0.646 ± 0.040	0.760 ± 0.052	0.564 ± 0.021	–	–	–
Parkinsons	0.766 ± 0.041	0.556 ± 0.036	0.669 ± 0.012	**0.847 ± 0.012**	0.726	0.731 ± 0.059	0.603 ± 0.128	0.564 ± 0.020	0.724 ± 0.019	0.520 ± 0.018	–	–	–
Obesity	0.843 ± 0.054	0.775 ± 0.035	0.747 ± 0.050	0.816 ± 0.008	**0.850**	0.836 ± 0.053	0.718 ± 0.200	0.695 ± 0.033	0.792 ± 0.019	0.578 ± 0.006	–	–	–
Diabetes	0.385 ± 0.023	0.391 ± 0.005	0.399 ± 0.009	**0.404 ± 0.014**	–	0.396 ± 0.016	0.336 ± 0.060	0.388 ± 0.022	0.388 ± 0.025	0.272 ± 0.011	0.374	–	–
German Credit	0.715 ± 0.129	0.761 ± 0.019	0.557 ± 0.073	**0.821 ± 0.007**	–	0.816 ± 0.015	0.732 ± 0.022	0.785 ± 0.023	0.750 ± 0.014	0.542 ± 0.010	–	–	–
Bank Marketing	0.704 ± 0.005	**0.709 ± 0.017**	0.690 ± 0.024	0.702 ± 0.014	–	0.648 ± 0.100	0.511 ± 0.111	0.586 ± 0.116	0.614 ± 0.053	0.404 ± 0.004	–	–	–
Census Income	0.518 ± 0.085	0.590 ± 0.026	**0.607 ± 0.044**	0.597 ± 0.060	–	0.541 ± 0.089	0.402 ± 0.289	0.531 ± 0.106	0.502 ± 0.049	0.305 ± 0.002	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Melanoma	0.341 ± 0.124	0.310 ± 0.003	0.288 ± 0.053	**0.387 ± 0.007**	0.384 ± 0.005	0.352 ± 0.049	0.285 ± 0.119	0.317 ± 0.033	0.376 ± 0.007	0.318 ± 0.006	–	–	–
PBC	0.447 ± 0.127	0.476 ± 0.192	0.450 ± 0.236	0.519 ± 0.276	**0.527 ± 0.270**	0.472 ± 0.224	0.445 ± 0.228	0.390 ± 0.086	0.500 ± 0.280	0.404 ± 0.150	–	–	–
PBC3	0.431 ± 0.184	0.416 ± 0.182	0.320 ± 0.020	0.480 ± 0.219	0.416 ± 0.226	0.434 ± 0.227	0.354 ± 0.198	0.378 ± 0.205	0.452 ± 0.198	0.344 ± 0.158	**0.545**	–	–
Retinopathy	0.768 ± 0.324	0.649 ± 0.071	0.611 ± 0.103	0.735 ± 0.038	**0.881 ± 0.015**	0.763 ± 0.106	0.606 ± 0.073	0.618 ± 0.016	0.720 ± 0.021	0.623 ± 0.030	–	–	–
WHAS500	0.516 ± 0.068	0.462 ± 0.069	0.451 ± 0.021	**0.560 ± 0.035**	0.524 ± 0.043	0.534 ± 0.018	0.424 ± 0.078	0.467 ± 0.061	0.530 ± 0.044	0.419 ± 0.030	–	–	–
GBSG2	0.671 ± 0.222	0.613 ± 0.097	0.472 ± 0.049	0.717 ± 0.140	**0.761 ± 0.161**	0.605 ± 0.208	0.562 ± 0.115	0.457 ± 0.120	0.720 ± 0.110	0.437 ± 0.135	0.669	–	–
Colon	0.876 ± 0.360	0.728 ± 0.030	0.696 ± 0.080	0.862 ± 0.013	**0.911 ± 0.027**	0.864 ± 0.065	0.669 ± 0.094	0.842 ± 0.007	0.827 ± 0.105	0.634 ± 0.037	–	–	–
GRACE	0.381 ± 0.017	0.337 ± 0.021	0.291 ± 0.024	**0.400 ± 0.005**	0.398 ± 0.006	0.340 ± 0.048	0.265 ± 0.200	0.350 ± 0.012	0.379 ± 0.026	0.316 ± 0.010	–	–	–
TRACE	0.403 ± 0.028	0.368 ± 0.028	0.331 ± 0.048	0.418 ± 0.005	**0.421 ± 0.001**	0.381 ± 0.018	0.330 ± 0.025	0.347 ± 0.099	0.392 ± 0.016	0.307 ± 0.006	–	–	–
Framingham	0.898 ± 0.008	0.678 ± 0.041	0.855 ± 0.016	0.920 ± 0.003	**0.927 ± 0.004**	0.867 ± 0.060	0.547 ± 0.021	0.692 ± 0.174	0.751 ± 0.053	0.600 ± 0.025	–	–	–
Blood Transfusion	0.610 ± 0.344	0.512 ± 0.070	0.547 ± 0.053	0.713 ± 0.014	**0.736 ± 0.015**	0.608 ± 0.043	0.474 ± 0.129	0.561 ± 0.056	0.643 ± 0.065	0.427 ± 0.026	–	–	–
Credit-G	0.667 ± 0.044	0.850 ± 0.033	0.611 ± 0.079	0.914 ± 0.020	**0.940 ± 0.016**	0.899 ± 0.057	0.748 ± 0.202	0.871 ± 0.056	0.856 ± 0.019	0.612 ± 0.003	–	–	–
QSAR Biodeg	0.707 ± 0.050	0.653 ± 0.014	0.686 ± 0.009	0.754 ± 0.026	0.576 ± 0.011	**0.768 ± 0.010**	0.550 ± 0.116	0.694 ± 0.024	0.712 ± 0.028	0.475 ± 0.013	–	–	–
Maternal Risk	0.746 ± 0.100	0.596 ± 0.043	0.505 ± 0.039	0.722 ± 0.028	0.657 ± 0.100	0.689 ± 0.112	0.539 ± 0.194	0.573 ± 0.005	**0.782 ± 0.021**	0.537 ± 0.045	–	–	–
Website Phishing	0.852 ± 0.243	0.765 ± 0.022	0.713 ± 0.012	0.796 ± 0.021	**0.965 ± 0.016**	0.910 ± 0.025	0.692 ± 0.099	0.902 ± 0.031	0.886 ± 0.005	0.750 ± 0.009	–	–	–

Table 28: Column shapes (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	0.748 ± 0.124	0.738 ± 0.042	0.614 ± 0.032	0.846 ± 0.020	0.681	0.773 ± 0.019	0.595 ± 0.120	0.696 ± 0.033	0.830 ± 0.041	0.634 ± 0.029	0.815	**0.875**	–
Breast Cancer	0.756 ± 0.052	0.649 ± 0.016	0.724 ± 0.082	0.868 ± 0.015	0.631	0.782 ± 0.028	0.255 ± 0.287	0.550 ± 0.060	**0.882 ± 0.039**	0.495 ± 0.027	–	–	–
Liver Disorders	0.643 ± 0.117	0.487 ± 0.106	0.612 ± 0.083	0.771 ± 0.011	**0.772**	0.632 ± 0.018	0.601 ± 0.179	0.535 ± 0.082	0.731 ± 0.068	0.434 ± 0.052	–	–	–
Parkinsons	0.773 ± 0.044	0.591 ± 0.048	0.700 ± 0.027	**0.863 ± 0.015**	0.676	0.714 ± 0.066	0.669 ± 0.236	0.522 ± 0.048	0.849 ± 0.016	0.522 ± 0.035	–	–	–
Obesity	0.851 ± 0.016	0.856 ± 0.045	0.821 ± 0.027	**0.893 ± 0.011**	0.889	0.885 ± 0.003	0.794 ± 0.185	0.848 ± 0.024	0.879 ± 0.007	0.664 ± 0.013	–	–	–
Diabetes	0.770 ± 0.046	0.782 ± 0.009	0.798 ± 0.017	**0.808 ± 0.028**	–	0.791 ± 0.032	0.672 ± 0.119	0.776 ± 0.044	0.776 ± 0.050	0.543 ± 0.022	0.748	–	–
German Credit	0.799 ± 0.052	0.872 ± 0.017	0.639 ± 0.011	**0.925 ± 0.007**	–	0.885 ± 0.019	0.817 ± 0.061	0.870 ± 0.016	0.854 ± 0.020	0.622 ± 0.009	–	–	–
Bank Marketing	**0.856 ± 0.020**	0.852 ± 0.019	0.812 ± 0.023	0.852 ± 0.021	–	0.809 ± 0.085	0.667 ± 0.139	0.767 ± 0.041	0.783 ± 0.052	0.515 ± 0.007	–	–	–
Census Income	0.663 ± 0.088	0.796 ± 0.057	0.782 ± 0.031	**0.818 ± 0.109**	–	0.726 ± 0.076	0.572 ± 0.403	0.730 ± 0.102	0.710 ± 0.102	0.426 ± 0.008	–	–	–
Melanoma	0.682 ± 0.247	0.620 ± 0.005	0.576 ± 0.107	**0.773 ± 0.014**	0.767 ± 0.009	0.703 ± 0.097	0.569 ± 0.237	0.634 ± 0.066	0.752 ± 0.013	0.637 ± 0.013	–	–	–
PBC	0.767 ± 0.042	0.790 ± 0.017	0.704 ± 0.032	0.823 ± 0.038	**0.837 ± 0.012**	0.767 ± 0.038	0.712 ± 0.053	0.744 ± 0.213	0.777 ± 0.025	0.679 ± 0.016	–	–	–
PBC3	0.781 ± 0.060	0.736 ± 0.053	0.640 ± 0.041	**0.859 ± 0.014**	0.737 ± 0.052	0.766 ± 0.031	0.619 ± 0.190	0.661 ± 0.043	0.819 ± 0.028	0.613 ± 0.008	0.818	–	–
Retinopathy	0.786 ± 0.090	0.814 ± 0.034	0.626 ± 0.047	0.841 ± 0.016	**0.850 ± 0.004**	0.794 ± 0.063	0.747 ± 0.078	0.795 ± 0.017	0.794 ± 0.016	0.768 ± 0.001	–	–	–
WHAS500	0.826 ± 0.083	0.770 ± 0.068	0.712 ± 0.023	**0.904 ± 0.004**	0.855 ± 0.032	0.855 ± 0.039	0.725 ± 0.145	0.767 ± 0.057	0.880 ± 0.016	0.689 ± 0.016	–	–	–
GBSG2	0.771 ± 0.154	0.710 ± 0.072	0.566 ± 0.031	0.808 ± 0.003	**0.831 ± 0.021**	0.687 ± 0.205	0.635 ± 0.242	0.701 ± 0.024	0.824 ± 0.030	0.512 ± 0.023	0.747	–	–
Colon	0.799 ± 0.582	0.823 ± 0.015	0.619 ± 0.065	0.872 ± 0.011	**0.879 ± 0.005**	0.754 ± 0.118	0.699 ± 0.154	0.815 ± 0.022	0.838 ± 0.021	0.630 ± 0.017	–	–	–
GRACE	0.761 ± 0.033	0.675 ± 0.041	0.582 ± 0.047	**0.800 ± 0.010**	0.797 ± 0.013	0.679 ± 0.097	0.531 ± 0.400	0.700 ± 0.024	0.757 ± 0.053	0.632 ± 0.020	–	–	–
TRACE	0.805 ± 0.056	0.735 ± 0.056	0.663 ± 0.096	0.837 ± 0.010	**0.843 ± 0.003**	0.763 ± 0.035	0.660 ± 0.049	0.695 ± 0.198	0.783 ± 0.031	0.613 ± 0.011	–	–	–
Framingham	0.806 ± 0.034	0.723 ± 0.079	0.726 ± 0.031	0.844 ± 0.012	**0.860 ± 0.003**	0.759 ± 0.058	0.494 ± 0.018	0.615 ± 0.313	0.768 ± 0.049	0.599 ± 0.016	–	–	–
Blood Transfusion	0.579 ± 0.307	0.539 ± 0.148	0.483 ± 0.058	0.695 ± 0.031	**0.732 ± 0.017**	0.531 ± 0.006	0.434 ± 0.207	0.576 ± 0.070	0.694 ± 0.052	0.370 ± 0.023	–	–	–
Credit-G	0.792 ± 0.018	0.921 ± 0.040	0.672 ± 0.041	**0.971 ± 0.008**	0.958 ± 0.018	0.918 ± 0.056	0.823 ± 0.180	0.919 ± 0.006	0.911 ± 0.022	0.654 ± 0.008	–	–	–
QSAR Biodeg	0.789 ± 0.064	0.772 ± 0.022	0.777 ± 0.021	0.791 ± 0.030	0.561 ± 0.034	0.849 ± 0.029	0.574 ± 0.167	0.775 ± 0.019	**0.873 ± 0.005**	0.417 ± 0.015	–	–	–
Maternal Risk	0.779 ± 0.023	0.663 ± 0.019	0.630 ± 0.029	0.764 ± 0.038	0.730 ± 0.073	0.732 ± 0.233	0.588 ± 0.268	0.724 ± 0.031	**0.867 ± 0.044**	0.585 ± 0.011	–	–	–
Website Phishing	0.908 ± 0.097	0.766 ± 0.016	0.686 ± 0.018	0.782 ± 0.009	**0.976 ± 0.010**	0.936 ± 0.032	0.769 ± 0.094	0.953 ± 0.033	0.920 ± 0.012	0.841 ± 0.014	–	–	–

Table 29: Column-pair trends (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000**	**0.000**	–
Breast Cancer	0.658 ± 0.015	0.550 ± 0.009	0.714 ± 0.017	**0.830 ± 0.015**	0.777	0.784 ± 0.021	0.553 ± 0.016	0.678 ± 0.008	0.611 ± 0.039	0.549 ± 0.010	–	–	–
Liver Disorders	0.967 ± 0.044	0.707 ± 0.008	0.798 ± 0.098	**0.969 ± 0.006**	0.959	0.942 ± 0.060	0.668 ± 0.035	0.757 ± 0.004	0.790 ± 0.036	0.694 ± 0.018	–	–	–
Parkinsons	0.759 ± 0.038	0.520 ± 0.025	0.637 ± 0.017	**0.831 ± 0.009**	0.777	0.749 ± 0.052	0.537 ± 0.028	0.606 ± 0.030	0.599 ± 0.042	0.518 ± 0.008	–	–	–
Obesity	**0.834 ± 0.099**	0.695 ± 0.026	0.673 ± 0.074	0.738 ± 0.008	0.812	0.788 ± 0.104	0.642 ± 0.217	0.541 ± 0.070	0.705 ± 0.037	0.492 ± 0.010	–	–	–
Diabetes	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	–
German Credit	0.630 ± 0.310	0.650 ± 0.033	0.475 ± 0.151	0.716 ± 0.020	–	**0.746 ± 0.011**	0.647 ± 0.045	0.701 ± 0.059	0.646 ± 0.039	0.462 ± 0.015	–	–	–
Bank Marketing	0.552 ± 0.017	0.566 ± 0.016	**0.568 ± 0.027**	0.553 ± 0.014	–	0.488 ± 0.116	0.356 ± 0.117	0.406 ± 0.192	0.445 ± 0.054	0.293 ± 0.007	–	–	–
Census Income	0.373 ± 0.083	0.383 ± 0.025	**0.431 ± 0.062**	0.375 ± 0.027	–	0.355 ± 0.103	0.233 ± 0.177	0.333 ± 0.124	0.293 ± 0.011	0.185 ± 0.003	–	–	–
Melanoma	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	–	–	–
PBC	0.127 ± 0.289	0.163 ± 0.392	0.197 ± 0.484	0.215 ± 0.544	0.217 ± 0.539	0.178 ± 0.449	0.178 ± 0.505	0.035 ± 0.076	**0.222 ± 0.559**	0.129 ± 0.306	–	–	–
PBC3	0.082 ± 0.353	0.096 ± 0.415	0.000 ± 0.000	0.101 ± 0.434	0.094 ± 0.404	0.102 ± 0.439	0.090 ± 0.386	0.096 ± 0.411	0.086 ± 0.368	0.075 ± 0.324	**0.272**	–	–
Retinopathy	0.751 ± 0.558	0.485 ± 0.108	0.595 ± 0.237	0.628 ± 0.087	**0.911 ± 0.034**	0.733 ± 0.153	0.465 ± 0.103	0.440 ± 0.024	0.646 ± 0.036	0.479 ± 0.062	–	–	–
WHAS500	0.206 ± 0.067	0.155 ± 0.070	0.190 ± 0.057	**0.216 ± 0.073**	0.192 ± 0.057	0.214 ± 0.069	0.123 ± 0.023	0.166 ± 0.067	0.181 ± 0.074	0.148 ± 0.072	–	–	–
GBSG2	0.572 ± 0.326	0.515 ± 0.144	0.378 ± 0.070	0.626 ± 0.278	**0.692 ± 0.339**	0.522 ± 0.234	0.489 ± 0.109	0.213 ± 0.264	0.616 ± 0.192	0.362 ± 0.256	0.591	–	–
Colon	0.954 ± 0.138	0.634 ± 0.056	0.772 ± 0.096	0.852 ± 0.036	0.943 ± 0.050	**0.974 ± 0.064**	0.639 ± 0.044	0.869 ± 0.017	0.816 ± 0.201	0.638 ± 0.083	–	–	–
GRACE	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
TRACE	**-0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	–	–	–
Framingham	0.991 ± 0.018	0.634 ± 0.033	0.985 ± 0.009	**0.996 ± 0.006**	0.995 ± 0.005	0.975 ± 0.071	0.600 ± 0.061	0.769 ± 0.040	0.734 ± 0.093	0.602 ± 0.040	–	–	–
Blood Transfusion	0.642 ± 0.408	0.484 ± 0.011	0.610 ± 0.054	0.731 ± 0.006	**0.740 ± 0.016**	0.685 ± 0.085	0.514 ± 0.076	0.547 ± 0.101	0.593 ± 0.129	0.484 ± 0.040	–	–	–
Credit-G	0.541 ± 0.069	0.779 ± 0.029	0.549 ± 0.119	0.858 ± 0.034	**0.921 ± 0.019**	0.879 ± 0.061	0.673 ± 0.228	0.823 ± 0.119	0.801 ± 0.022	0.569 ± 0.006	–	–	–
QSAR Biodeg	0.625 ± 0.036	0.534 ± 0.037	0.595 ± 0.020	**0.716 ± 0.065**	0.591 ± 0.040	0.688 ± 0.042	0.526 ± 0.088	0.613 ± 0.039	0.551 ± 0.051	0.532 ± 0.035	–	–	–
Maternal Risk	**0.712 ± 0.178**	0.528 ± 0.090	0.381 ± 0.051	0.681 ± 0.037	0.584 ± 0.133	0.645 ± 0.014	0.491 ± 0.137	0.422 ± 0.023	0.698 ± 0.068	0.490 ± 0.079	–	–	–
Website Phishing	0.796 ± 0.390	0.765 ± 0.035	0.739 ± 0.036	0.809 ± 0.037	**0.955 ± 0.021**	0.883 ± 0.021	0.616 ± 0.106	0.851 ± 0.031	0.851 ± 0.005	0.660 ± 0.021	–	–	–

Other

Table 30: Parse valid-rate (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN
Heart Failure	**0.675 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Breast Cancer	**0.693 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Liver Disorders	**0.680 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Parkinsons	**0.655 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Obesity	**0.708 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Diabetes	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
German Credit	**0.537 ± 0.708**	–	–	–	–	–	–	–	–	–	–	–	–
Bank Marketing	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Census Income	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Melanoma	**0.659 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
PBC	**0.474 ± 0.873**	–	–	–	–	–	–	–	–	–	–	–	–
PBC3	**0.680 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Retinopathy	**0.682 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
WHAS500	**0.688 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
GBSG2	**0.696 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Colon	**0.700 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
GRACE	**0.702 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
TRACE	**0.708 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Framingham	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Blood Transfusion	**0.529 ± 0.701**	–	–	–	–	–	–	–	–	–	–	–	–
Credit-G	**0.702 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
QSAR Biodeg	**0.703 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Maternal Risk	**0.688 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–
Website Phishing	**0.698 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–